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Manual



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1. **Introduction**

This style manual is intended primarily for use by WHO editors working in English. Its main purpose is to ensure that the language used in WHO publications is clear and correct and that the scientific and other special nomenclature and terminology, abbreviations and symbols are reasonably consistent and follow established international precedents.

Editors should treat the manual as a work of reference to be consulted in case of doubt, and not as a standard against which all details of every manuscript must routinely be matched. Time should not be spent in the pursuit of uniformity as an end in itself. The manual is intended to speed up the preparation of manuscripts, not retard it.

Other WHO staff members who write articles or reports for eventual publication by WHO will greatly facilitate the work of the editors if they follow the conventions established here. However, the manual is not intended to be a guide to English grammar or to good writing: readers seeking such a guide are referred to *The elements of style*, 3rd ed. (Strunk W, White EB. New York, Macmillan, 1978) and *A guide to writing for the United Nations* (New York, United Nations, 1984).

The manual has been prepared by the Technical Publications subunit, Office of Publications (PUB/TPS). It will be updated as necessary and revised in the light of experience with its use. Comments on the content of the manual and suggestions for changes or additions are welcome and should be sent to PUB/TPS.

2. Spelling

This section is a general guide to spelling used in WHO publications, and also gives guidelines on hyphenation and capitalization. It is not intended to be comprehensive, but to cover the main areas, such as scientific and medical terminology, and names of places and organizations, where confusion and uncertainty may arise.

Within WHO, British rather than American spelling is normally used. The general rule is to follow the first spelling listed in the latest edition of *The concise Oxford dictionary*. If a variant spelling is listed separately, the one to which you are referred should be used; e.g. “tire, US var. of tyre”: use “tyre”.

There are, however, a number of exceptions:

- where a different spelling has become established usage in WHO;
- where there is a need to respect the recommendations of international nomenclature-setting bodies;
- quoted material, where the original spelling must be reproduced exactly; and
- book and article titles, where the original spelling must be reproduced.

Annex 1 gives the preferred spellings of many of the commonly encountered exceptions; it also includes *The concise Oxford dictionary* spellings that should be followed for many words that often cause doubt or difficulty, and gives some guidelines on usage and choice of words.

For transliteration from Arabic, Russian and Chinese see section 7.

Medical and scientific terminology

The spelling of disease names and other medical terms follows British rather than American usage. Some frequently found exceptions are:

estrogen, estrus, etc. (but oesophagus and oedema)
etiology
fetal, fetus, etc.
leukocyte, leukopenia.

For disease names and their spelling, the International Nomenclature of Diseases (IND) should be followed. This must not be confused with the International Classification of Diseases (ICD), which is a statistical classification and not a nomenclature.

Eponymous names give no information about the nature of the disease and different names may be used in different countries. They should be avoided whenever possible. The Saxon genitive ('s) is not necessary when using an eponymous name; e.g. Crohn disease, not Crohn's disease, and Down syndrome, not Down's syndrome.

For general reference, *Butterworths medical dictionary*, 2nd ed. (Critchley M, ed. London, Butterworths, 1978) and *Dorland's illustrated medical dictionary*, 28th ed. (Philadelphia, Saunders, 1994) can be recommended for British and American spelling, respectively.

Progress in medical terminology (Manuila A, ed. Basel, Karger, 1981) describes the evolution of medical terminology and gives a detailed account of the development of the IND.

For scientific spelling and terminology see section 14.

Country and place names

Webster's new geographical dictionary (Springfield, Merriam-Webster, 1984) can be used as a guide to the spelling of common geographical entities. However, country names and their spelling, in particular, can change at an alarming rate. The Computer-Assisted Translation and Terminology subunit (CTT), which coordinates with the Office of the Legal Counsel (LEG), can provide up-to-date information on the names of United Nations and WHO Member States, including their full, short and adjectival/attributive forms.

The official names of United Nations Member States and approved and alternative names for various cities and countries are listed in Annexes 2 and 3. For countries such as "The Gambia" and "The Netherlands", the definite article may be omitted in address lists and enumerations.

Syllabification

The first edition of *Collins dictionary of the English language* (Hanks P, ed. London, Collins, 1979) gives syllabification breaks and appropriate hyphenation points for each entry. The position of syllabic breaks where hyphenation is permissible is based upon phonetic rather than etymological principles, e.g. cata-lyse, but ca-taly-sis; meta-bol-ic, but met-abo-lism.

Hyphenation

Hyphens are used to connect words that are more closely linked to each other than to the surrounding syntax. Unfortunately there are no hard and fast rules about their use. Certain conventions exist, but if there is a choice it is better to introduce hyphens only to avoid ambiguity. There is an overall tendency to use hyphens less than in the past and a few general rules are applicable, as follows.

Hyphen after a prefix

A hyphen should be used after a prefix in order to prevent the word being mistaken for another one, e.g. co-op, coop (but cooperate needs no hyphen); re-cover, recover; un-ionized, unionized.

Prefixes such as anti- and semi- can often be used without a hyphen, e.g. antihypertensive, antidysrhythmic, semiconductor, semicircular. However, a hyphen is required to avoid doubling a vowel or a consonant, e.g. anti-inflammatory, anti-infective, semi-invalid.

A hyphen is also required when the prefix is to a word beginning with a capital letter, e.g. anti-Darwinian, sub-Saharan (but note Pan American Health Organization).

Hyphens with compound nouns and adjectives

Some compound words remain hyphenated irrespective of their grammatical use, e.g. well-being, side-effect, extra-articular, forty-two, Vice-President, Director-General. Others are hyphenated according to whether they are used as nouns or adjectives, e.g. first-class (adj.) *but* first class (n.), low-cost (adj.) *but* low cost (n.), or whether they are used attributively or as predicates, e.g. up-to-date statistics *but* the statistics are up to date.

It is usually unnecessary to insert a hyphen after an adverb, e.g. carefully developed plan, well known fact, but a hyphen is appropriate when an adverb might be mistaken for an adjective (compare little-used car and little used car).

WHO usage

The following is a list of hyphenated terms that appear commonly in WHO publications:

above-mentioned	enzyme-linked immunosorbent assay
after-care	evaporating-dish
after-effect	extra-articular
air-conditioning	
anti-gas-gangrene	fact-finding
anti-infective	false-positive reaction
anti-inflammatory	far-reaching
	field-test (vb.)
blood-grouping	filter-paper
blood-meal	fine-tooth comb
blood-typing	finger-prick (n. & adj.)
boiling-point	first-aid kit
book-keeper	flow-chart
bottle-feed (vb)	follow-up (n. & adj.)
bottle-feeding	food-chain
breast-milk substitute	foot-rule
by-law	freezing-point
by-product	front-line (adj.)
case-finding	gas-gangrene
case-load	Gram-positive/negative
case-study	grass-roots (adj.)
check-up (n.)	guinea-pig
child-spacing	guinea-worm disease
co-eluting	
cold-chain equipment	half-hour
common-sense (adj.)	half-yearly
co-trimoxazole	health-for-all (adj.)
co-worker	hydrogen-ion concentration
cross-matching	
cross-react, cross-reaction	ill-health
cross-resistance	in-service training
cross-section	inter-American
cut-off point	intra-abdominal
decision-maker/making	juxta-articular
Director-General	
drinking-water (but International Drinking Water Supply and Sanitation Decade)	knee-jerks
drop-out (n.)	know-how
end-point	labour-intensive
end-product	<i>laissez-faire</i>
	<i>laissez-passer</i> (inv. sing. & pl.)
	large-scale (adj.)

last-mentioned	seat-belt
life-support (adj.)	self-care
long-distance (adj.)	semi-arid
long-sighted (adj.)	set-back (n.)
long-sightedness (n.)	set-up (n.)
long-term (adj.)	shelf-life
louse-borne	short-sighted (adj.)
low-cost (adj.)	short-sightedness (n.)
lump-sum reimbursement	short-term (adj.)
lying-in hospital (n.)	sick-leave
man-hours/months, etc. (<i>prefer</i>	sickle-cell anaemia
work-hours/months, etc.)	side-effects
man-made	small-scale (adj.)
needle-stick (n. & adj.)	snake-bite
non-profit-making	so-called (adj.)
non-self-governing	South-East Asia Region (WHO)
non-wage-earning	south-east, south-eastern
north-east, north-eastern	south-west, south-western
north-west, north-western	starting-point
one-third	strait-jacket
out-of-date (adj.)	strait-laced
policy-maker, policy-making	sub-Saharan
post-session	sub-Saharan
post-test	sugar-beet
post-treatment	sugar-cane
précis-writer	test-tube
pre-eclampsia	tick-borne
pre-eminent	time-limit
pre-empt, pre-emption,	time-scale
pre-emptive	two-thirds
pre-session	ultra-low-volume formulations
primary-school child(ren)	(abbr.= ULV)
re-edit	up-to-date (adj.)
re-educate	vector-borne
re-employ	vice-chairman, etc.
re-enter, re-entry	<i>vis-à-vis</i>
re-establish	wage-earner, wage-earning
re-examine	water-bath
re-form (form again)	well-baby clinic
re-route	well-being
re-treat (treat again)	well-water
roll-call	work-hours/months, etc.
	X-ray

Capitalization

Too many capital letters break the flow of the eye across a sentence and down a column of type. However, proper names, titles and institutions require capitals. Descriptive appellations, such as administrative posts, do not, unless the result would appear odd or confusing.

The trade names of drugs and other proprietary products, e.g. Pyrex and Vaseline (petroleum jelly preferred), require an initial capital letter. Generic terms in the Linnaean classification also take an initial capital letter, e.g. *Staphylococcus aureus*, *Trypanosoma*, *Schistosoma*, but trypanosome, schistosome, etc.

Article, book and journal titles in references need not be capitalized, e.g. *Social science and medicine*, except when they contain the name of an association or organization, e.g. *Bulletin of the World Health Organization*, *Journal of the American Medical Association*.

The following words and expressions used frequently within WHO normally require capitalization:

- Associate Member
- Constitution (of WHO)
- (Eighth) General Programme of Work
- Forty... World Health Assembly
- Global Strategy for Health for All by the Year 2000
- Member (country)
- Member State
- officials (specific): President, Vice-President, Chairman, Rapporteur, Secretary
- the Organization (WHO)
- Region [of WHO] (cap. in specific reference, e.g. "in the European Region")
- Regional Committee (specific)
- Regional Director(s) (specific)
- Regional Office (specific), but regional offices (pl.)
- Secretariat (WHO)

Some other commonly used words are capitalized only when used specifically, e.g. a government, but the French Government; a ministry, but the Ministry of Health; a department, but the Department of the Environment.

The following words and expressions do not normally require capitalization:

- commission, committee (unless specifically identified)
- decision (e.g. ... in decision EB62(4) ...)
- health for all
- headquarters (WHO)
- member (of the committee, etc.), but Member (State)
- names of seasons
- ordinal numbers referring to sessions of the Board (e.g. sixty-third session of the Executive Board)
- primary health care
- programme (except those established by a World Health Assembly resolution, e.g. Global Programme on AIDS, Expanded Programme on Immunization)
- region(s) (when used generically)
- representatives
- resolution (e.g. ...in resolution WHA41.26 ...)
- session
- specialized agency
- strategy, strategies (e.g. health-for-all strategy)
- twenty-first century

For further information on capitalization see *Hart's rules for compositors and readers at the University Press Oxford*, 39th ed. (Oxford, Oxford University Press, 1983:8–14).

Words ending in -ize, -ise and -yse

Where there is a choice between using the suffix *-ize* or *-ise* (e.g. organize or organise), *-ize*, derived from the Greek “-izo”, is preferred, consistent with the first spelling of such words given in *The concise Oxford dictionary*. For some words, such as surprise, comprise or improvise, where *-ise* is not a suffix but part of the root of the word, there is no choice and *-ise* must be used. Similarly, *-yse* must be used for words derived from the Greek *lusi* (= lysis), as in analyse, dialyse and hydrolyse.

Alphabetical arrangement

In lists of names and addresses, indexes, alphabetical reference lists, etc., the following general rules should be followed:

- When the first component of a family name is a particle such as “de”, “Le”, “al-”, “Van” or “von”, the first letter of the particle should normally be used for alphabetization, unless established usage or another specific reason dictates otherwise.
- Names beginning with the prefix “Mac” or “Mc” should be arranged as if the letter “a” were present in all cases.
- When Chinese personal names are written in the traditional way, the single-syllable family name comes first and should be used for alphabetizing (e.g. Dr Hu Ching-Li should be listed under “H”).
- In the alphabetization of chemical names, Greek letters and italicized prefixes should be ignored.

3. *Punctuation*

This is a large subject where personal taste and common sense are more important than any rules. Appropriate punctuation will ease reading and clarify meaning, and this should be its purpose.

A well written running text should require only the minimum of punctuation in order to read clearly and sensibly. If much punctuation is required, it is probably a good indication that the text has been badly written and should be revised rather than heavily punctuated. Particularly in scientific writing, unnecessary punctuation should be avoided if the meaning is clear.

The punctuation guidelines below do not cover the special requirements of references and bibliographies (see section 9), or those of indexes.

Care should be taken to make sure that, when appropriate, punctuation marks are printed in the same style and type font as the rest of the text in which they appear.

Very few specific examples of the use of punctuation marks appear in this section, since they are to be found in abundance in the books listed on page 15.

Full point (period)

A full point should be used to separate statements between which there is no true continuity of thought. This results in shorter, more concise sentences, thus avoiding the temptation to include too much in a single sentence.

No full point is required for column headings, running heads, chapter titles and legends, or after abbreviations ending with the final letter of the word (see "Abbreviations and symbols", section 11, for examples).

An abbreviating full point, as at the end of etc. and Co., is omitted before a full point ending a sentence. Similarly, a stronger punctuation mark, such as a question mark or an exclamation mark, will replace a full point at the end of a sentence.

Ellipsis points (three spaced dots) signify either omitted material or faltering speech. Interruptions or abrupt changes in thought should be indicated by a dash (em rule).

For an explanation of the correct use of the full point in quotations see section 6.

Comma

Correct usage of the comma is often a question of judgement. Commas are now used much less frequently than in the past, and their main purpose is to indicate a pause or to avoid ambiguity.

In a list of three or more items, a comma should in general be consistently inserted or omitted before the final “and”, e.g.

X, Y and Z *or* X, Y, and Z.

Editors should ensure that each publication is internally consistent in this respect. However, even when the “X, Y and Z” system is adopted, an additional comma should be inserted wherever necessary to avoid ambiguity, e.g.

W, X and Y, and Z *differs in meaning from* W, X, and Y and Z.

Commas can often be used instead of parentheses and when separating relative clauses within a sentence.

It is usual to omit the comma when printing house numbers in addresses.

Semicolon

The semicolon marks a longer pause or more definite break than the comma. It links clauses or sentences too closely related to be cut off by a full point, but not closely enough related to be joined by a conjunction.

The semicolon is used to separate two or more linked clauses of more or less equal importance.

In a sentence containing long clauses with commas, the semicolon is used to indicate major breaks or appropriate pauses.

The first word following a semicolon always starts with a lower-case letter, unless it is a proper noun.

Colon

The systematic use of the colon as one of a series of different pause-values has almost died out. It has three important uses:

1. To mark the antithesis between two statements more sharply than a semicolon.
2. To introduce a list or series – never followed by a dash.
3. To indicate that the second statement is an explanation or amplification of the first.

Parenthesis

A parenthesis can be marked off by commas, dashes or brackets, depending upon the closeness of its relationship to the sentence.

Parentheses should be used sparingly since they tend to break up sentence structure and can too often be used to save the trouble of organizing one's thoughts.

Spaced em or en rules are used as parenthetical dashes. If unspaced in the typescript, they should have a vertical line on either side, but a space sign is unnecessary.

Dashes

Hyphen

It is not possible to be didactic about hyphens without being misleading, but they should be avoided whenever possible and introduced only to avoid ambiguity.

The practice of separating two or more prefixes to the same word, leaving a hyphen in mid-air, should be avoided if possible, e.g.

... hypertension or hypotension ... *not* ... hyper- or hypotension ...

Use of the hyphen with appropriate examples is described in greater detail in section 2.

En rule

An en rule is longer than a hyphen and is most commonly used to convey a distinction in sense when the first part of the compound does not modify the meaning of the second part – unlike the hyphen. The en rule in these circumstances can usually be thought of as standing for “and” or “to”, and is usually unspaced, e.g.

case–control study
benefit–risk ratio
oil–water interface
gas–liquid chromatography.

In the above examples a solidus (oblique stroke) would convey the same meaning, e.g. oil/water interface.

The en rule signifies “to” in the following:

12–24 September
pp. 5–55
London–Brighton race.

However, if the word “from” is used, e.g. from 1850 to 1895, then the en rule is inappropriate and “to” must be used.

For typographical reasons, a spaced en rule is sometimes used – as in this manual – to perform the functions of an em rule.

Em rule

The role of the em rule or dash in parentheses has been mentioned already, but it also has a number of other important uses:

1. To introduce an explanation or amplification of what immediately precedes it.
2. To gather up the subject of a long or complicated sentence.
3. To introduce a paradoxical or humorous ending to a sentence.
4. To indicate the omission of a word or part of a word.
5. To introduce items in a list (see section 10) or lines of dialogue, or to list phrases.

Brackets

The word “brackets” usually signifies square brackets; round brackets are “parentheses”, and curly brackets, sometimes used to group items in a table, are “braces” (see section 13 for their use in mathematical formulae).

Square brackets are used to indicate words interpolated in quotations; their contents do not affect the punctuation of the quotation. Similarly, square brackets may be used to enclose an explanation within the text by someone other than the author.

Question mark

Question marks are necessary at the end of direct but not of indirect questions.

Requests in question form for the sake of politeness, e.g. would you kindly tell me whether..., usually have question marks, although they are not strictly necessary.

Further reading

Many excellent examples of correct usage of the above punctuation marks, and much more besides, are to be found in the following:

Butcher J. *Copy-editing. The Cambridge handbook for authors, editors and indexers*, 3rd ed. Cambridge, Cambridge University Press, 1992.

Carey GV. *Mind the stop: brief guide to punctuation*. Harmondsworth, Penguin Books, 1971.

Council of Biology Editors Style Manual Committee. *Scientific style and format: the CBE manual for authors, editors, and publishers*, 6th ed. Cambridge, Cambridge University Press, 1994.

Gowers E (rev. Greenbaum S, Whitcut J). *The complete plain words*, 3rd ed. London, HMSO, 1986.

Hart's rules for compositors and readers at the University Press Oxford, 39th ed. Oxford, Oxford University Press, 1983.

Strunk W, White EB. *The elements of style*, 3rd ed. Macmillan, New York, 1978.

The Chicago manual of style: the essential guide for writers, editors, and publishers, 14th ed. Chicago, University of Chicago Press, 1993.

4. *Italics*

Italic type in printed publications is indicated in typescript by underlining. Italics are used in WHO publications in the following ways:

- for the titles of books, journals and documents;
- for foreign words and expressions not in common use (including Latin species and genus names);
- for variables in mathematical expressions;
- for certain letters, prefixes and terms in scientific use;
- (sparingly) for words requiring special emphasis.

Titles of books and journals

See “Reference lists and bibliographies”, section 9.

Foreign words and expressions

Italicize foreign words and phrases not in common use. The list below gives examples of current WHO practice in this regard; for words not in this list, consult the latest edition of *The concise Oxford dictionary*, or *Hart’s rules for compositors and readers at the University Press Oxford*, 39th ed. (Oxford, Oxford University Press, 1983). The names of institutions and organizations such as *Médecins sans frontières* should generally be italicized in running text but not in address lists.

<i>Italicize:</i>	<i>pro tempore</i>	corrigendum	lacuna
<i>a priori</i>	<i>raison d’être</i>	crèche	laissez-passer
<i>carte blanche</i>	<i>sensu stricto</i>	curriculum	per annum
<i>chef-d’oeuvre</i>	<i>sine qua non</i>	vitae	per capita
<i>coup d’état</i>	<i>vis-à-vis</i>	de facto	précis
<i>en masse</i>		e.g.	prima facie
<i>en route</i>	<i>Print in roman:</i>	élite	pro rata
<i>fait accompli</i>	addendum	erratum	rapporteur
<i>in situ</i>	ad hoc	et al.	résumé
<i>in vitro</i>	alias	etc.	status quo
<i>in vivo</i>	attaché	et seq.	verbatim
<i>laissez-faire</i>	avant-garde	ex officio	versus
<i>modus operandi</i>	blasé	habeas corpus	via
<i>per se</i>	bona fide	in camera	vice versa
<i>pièce de</i>	cliché	incommuni-	
<i>résistance</i>	communiqué	cado	

Variables in mathematical expressions

See “Mathematics and statistics”, section 13.

Scientific terms

See “Scientific terminology”, section 14.

Words requiring special emphasis

Use italics *sparingly* to indicate emphasis.

5. **Footnotes**

Footnotes are used to give explanatory or supplementary details when insertion in the text itself is not advisable, because the insertion would break the continuity of the text unduly, because of the length of the passage, or because the details were added by someone other than the author. Such details might include references, explanations of unfamiliar terms, last-minute additions or editor's notes.

The number of footnotes should be kept to a minimum. As a general rule, very short notes, such as cross-references to other sections or pages of the same article or work, should be given, in parentheses, in the text.

Complicated tabular material, lists and other entities that are not part of the text should be put in an annex at the back of the book, not included in footnotes. The footnote may read simply:

¹ For a list of institutions involved see Annex 1.

For footnotes to tables see section 10.

Identification

Footnotes to the text should always be placed at the bottom of the appropriate page. They are usually identified by superscript arabic numerals. When this may cause confusion (e.g. in scientific text containing mathematical exponents), lower-case letters should normally be used. Asterisks and other special typographical signs are used only rarely (e.g. to indicate levels of statistical significance in tables and, in membership lists for meetings of expert groups, to identify people who were invited but unable to attend).

Footnotes to the text should be numbered consecutively, beginning with "1", throughout each page of a book or article of a journal. If two (or more) passages appearing on the same page require the same footnote, one footnote only should be given for both, the reference mark being repeated in the text as appropriate.

If any punctuation immediately follows the word, part of a sentence, or sentence to which a footnote refers, the reference mark should be placed after the punctuation mark. The reference mark should be placed after a bracket if it refers to the text in parentheses; otherwise it should be placed immediately after the word or phrase to which it refers.

If a footnote refers to a quotation, the reference mark should be placed after the final quotation marks.

Editor's notes

Notes supplied by an editor, translator or compiler in a work including original notes by the author must be differentiated from the original notes. Such notes may be numbered in sequence with the author's notes.

Paragraphing and style

The paragraphing of footnotes should be in keeping with the overall design of the book. They may be set flush or in regular indented paragraph style. Each footnote should end with a full stop.

6. Quotations

All quotations from printed material should correspond exactly to the original in wording, spelling, punctuation, use of capital letters, etc. When part of the original text is omitted from a quotation, the passage in question should be replaced by three spaced dots. When three points are used at the end of an incomplete sentence a fourth full point should not be added. However, where the sentence is complete, the closing point is set close up, followed by three points for omission. Omissions should not change the meaning of the quotation.

Words inserted in or added to a quotation for clarity should be enclosed in square brackets.

Short quotations should be enclosed in quotation marks and incorporated in the body of the text. Long quotations (more than three lines) should usually be set off in some way (indented and/or smaller type). In this case, quotation marks are not needed.

When a quotation occurs within a quotation, double quotation marks should be used for the first quotation, single marks for the second one.

All signs of punctuation used with words in quotation marks should be placed according to the sense. Thus, if an *extract* ends with a point or exclamation or question mark, it should be included within the quotation marks. In such a case, no final full stop is needed outside the quotation marks. A number of useful examples are given in *Hart's rules for compositors and readers at the University Press Oxford*, 39th ed. (Oxford, Oxford University Press, 1983:44–49).

Normally, quotations from a foreign language should be translated into English, the words (translation from [the original language]) being inserted at the end of the quotation. If a quotation contains only a few foreign words, they may be left untranslated.

1

2

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7. Transliteration

It is usual to transliterate Arabic and Russian from their original alphabets to the Latin alphabet, and to transcribe Chinese into Latin characters, i.e. to romanize the language. Editors should consult the Office of Language Services (TRA) if they are in any doubt about the transliteration of particular words or phrases.

Arabic

The system of transliteration used by the United Nations is given in document ES/TRANS/84/3/rev 2, *Capitalization and spelling of some terms related to the Middle East and North Africa*. This document also contains useful information on the official names of various people, places and organizations often mentioned in connection with Arab affairs. A copy is kept in CTT. References in general texts to well known people or places should employ the spelling familiar to English-speaking readers, e.g. Mecca not Makka.

Note the following in relation to Arabic proper names:

- The word “al” often represents the definite article, in which case it is attached to the following word with a hyphen, and is lower case, e.g. Mr Hafez al-Assad.
- Abd (al-), “Servant (of the)” cannot be separated from the name that follows. A reference, for instance, to Mr Abd al-Qadir cannot subsequently be shortened to Mr Al-Qadir or Mr Qadir.

Russian

There are a number of systems for transliterating Russian. The ISO “linguistic system”, making free use of diacritical marks and with a one-to-one correspondence between letters, is recommended for use in WHO publications. The most recent version is ISO 9-1995: Information and documentation – Transliteration of Cyrillic characters into Latin characters – Slavic and non-Slavic languages. A slightly different older version (ISO/R9-1954) is currently in general use in WHO.

Regardless of the system used, well known Russian names should be given in the form familiar to English-speaking readers, e.g. Moscow, Tchaikovsky. Similarly, if an author or a participant in a meeting

expresses a preference for a particular spelling of his or her name, this should be respected, and an already-published spelling should not be changed without good reason.

Chinese

The romanization system called pinyin, introduced by the Chinese in the 1950s, has now largely supplanted the older Wade-Giles system. Pinyin should be used in all WHO publications; accents are not generally necessary, except in certain specific contexts (e.g. texts on acupuncture). Pinyin spellings often differ markedly from the older ones, and personal names are usually spelt without apostrophes. Given names tend to be closed up rather than hyphenated in pinyin, but current practice is variable. A list of Chinese place names, giving the pinyin spelling, the Han character name, and the usual English, French and Spanish equivalents is available from TPS.

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In a report of a meeting, a disclaimer along the following lines should be used (in Technical Reports, this appears on the title page):

This report contains the collective views of an international group of experts, and does not necessarily represent the decisions or the stated policy of the World Health Organization.

Disclaimers are intended to protect the Organization, but particular care should nevertheless be taken with the wording of text dealing with politically or legally sensitive issues (see section 16).

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9. *Reference lists and bibliographies*

A clear distinction should be made between lists of references and bibliographies. A list of references contains only those works cited in the text as sources of data or other information. It may include unpublished documents, provided that these are available to interested readers. A bibliography is a list of published works relevant to the subject matter of the publication and recommended for further reading.

Authors are responsible for ensuring the accuracy of bibliographic information. However, the editor should check any details that are obviously wrong or that appear doubtful.

Bibliographies

Bibliographies in WHO publications should be reasonably representative of the world literature on the subject. It should be remembered that most WHO books are translated into at least French and Spanish. The names of authors should be arranged alphabetically, but a bibliography may be subdivided into sections by subject. Bibliographies should not contain references to unpublished works.

References

References may be cited in the text using either a numerical system or the Harvard system. The numerical system is preferred, and is obligatory for the Technical Report Series, the *Bulletin of the World Health Organization* and *World health forum*. A reference to a finding reported in an unpublished document should simply be followed, in parentheses, by the statement “unpublished data” or “unpublished observations” accompanied, if appropriate, by the name of the authority cited and the date (P. Butler, unpublished data, 1991). Personal communications should be dealt with similarly. However, if an unpublished document is known to be easily available, it may be included in the reference list, together with an indication of how it may be obtained (see example on page 29).

Numerical system

The references are numbered consecutively as they occur in the text, the number being in italic (underlined in the manuscript) and placed

in parentheses. The authors' names may or may not be given:

Calvin & Hobbs (12) have reported ...

A recent study in India (3) showed ...

Harvard system

The references are cited by giving the names of the authors followed by the year of publication in the text. This may be done in one of two ways:

Ballance, Butler & Ewart (1989) have reported ...

It has been reported (Ballance, Butler & Ewart, 1989; Allsopp, 1990) that ...

Bibliographic style

Bibliographic style in WHO publications is based on the "Uniform requirements for manuscripts submitted to biomedical journals" (the so-called Vancouver style), formulated by the International Committee of Medical Journal Editors, but with certain adaptations to meet WHO's particular needs. (It should be noted in any case that the Vancouver style is intended to provide instructions to authors on how to prepare their manuscripts, and is not a style guide for editors.)

1. List all authors when three or fewer; when four or more, give only first author's name and add et al.
2. Write out journal names in full. This is particularly helpful to readers whose mother tongue is not English.
3. Use punctuation as given in the examples below.

Journal article

Feldmeier H, Doehring E, Little P. Clinical experience with metrifonate. *Acta tropica*, 1987, 44:357–368.

Book

Jordan P. *Schistosomiasis – the St Lucia project*, 2nd ed. Cambridge, Cambridge University Press, 1985.

Chapter in a book

Pawlowski ZS. Control of ascariasis within primary health care. In: Crompton DWT et al., eds. *Ascariasis and its public health significance*. London, Taylor & Francis, 1985:245–252.

Monograph in a series

Cancer pain relief and palliative care. Report of a WHO Expert Committee. Geneva, World Health Organization, 1990 (WHO Technical Report Series, No. 804).

WHO Expert Committee on Biological Standardization. *Fortieth Report.* Geneva, World Health Organization, 1990 (WHO Technical Report Series, No. 800).

Corporate author

Royal Marsden Hospital Bone-Marrow Transplantation Team. Failure of syngeneic bone-marrow graft without preconditioning in post-hepatitis marrow aplasia. *Lancet*, 1977, 2:242–244.

If the corporate author is also the publisher, it need only be mentioned once. Where a WHO publication has no named author, the World Health Organization should be shown as the publisher:

The international pharmacopoeia. Vol. 3. Quality specifications. Geneva, World Health Organization, 1988.

Dissertation or thesis

Cairns RB. *Infrared spectroscopic studies of solid oxygen* [Dissertation]. Berkeley, CA, University of California, 1965.

Published proceedings paper

DuPont B. Bone marrow transplantation in severe combined immunodeficiency with an unrelated MLC compatible donor. In: White HJ, Smith R, eds. *Proceedings of the third annual meeting of the International Society for Experimental Hematology.* Houston, TX, International Society for Experimental Hematology, 1974:44–46.

Unpublished document (if available to readers)

Guidelines for leishmaniasis control at regional and subregional level. Geneva, World Health Organization, 1988 (unpublished document WHO/LEISH/88.25; available on request from Division of Control of Tropical Diseases, World Health Organization, 1211 Geneva 27, Switzerland).

Foreign language reference

Jordan JR. Desarrollo psicomotor del niño. [Psychomotor development of the child.] In: *Temas de pediatría. [Aspects of paediatrics.]* Havana, Editora Universitaria, 1976.

List style

Whichever citation system has been used, the references should be listed in order (numerical or alphabetical) at the end of the chapter or full text, as appropriate, before any annexes.

If the Harvard system of citation has been used, the format shown in the above examples should be modified slightly, the date being placed in parentheses immediately after the authors' names:

Davis A, Bailey DR (1969) Metrifonate in urinary schistosomiasis. *Bulletin of the World Health Organization*, 41:209–224.

This system requires that each reference have a named author or be listed under Anon. It is preferable to list, for example, WHO publications without named authors under WHO. In all cases, the citation in the text should correspond exactly to the form used in the reference list.

In the Harvard system, references should be listed in alphabetical order, and the following rules should be observed:

- A single-author entry comes before a multiauthor entry beginning with the same name.
- All works attributed to one person should be listed together in chronological order, by year of publication.
- Two or more works by the same author and published in the same year are distinguished by letters after the date:

Clark S (1990a) ...

Clark S (1990b) ...

Typographical style

In WHO publications, titles of books and journals are always given in italics. Use of other fonts, e.g. bold for volume numbers or authors' names, will depend on the book design.

10. *Manuscript mark-up and layout*

Manuscript mark-up

For most manuscripts destined for publication by WHO, the typesetter will have a design specification prepared by the Graphics unit (GRA) or a previous book as a model to work from, so that detailed typographical mark-up is rarely necessary. It is generally sufficient for editors to:

- code headings (A, B, C etc.);
- identify ambiguous letters, signs and symbols (Greek characters, en rules etc.);
- ensure that requirements for bold, italic and small type and for capital and lower-case letters are clear;
- indicate the required format for lists, quotations and any other passages to be treated differently from the main text;
- note in the margin the approximate positions for insertion of figures and tables;
- clearly separate from the main text (either physically or by means of a coloured pen) figures, tables and their headings and footnotes;
- ensure that spacing requirements are unambiguous (insert a vertical line to fill up a large area of unwanted space and mark “run on” or “new page” wherever necessary).

In addition, all instructions for the printer and editorial comments (e.g. “check in proof”) should be circled so that it will be clear that they are not to be typeset.

For all publications, the preliminary pages prepared for the typesetter should include the title page, copyright page (with space left for Cataloguing in Publication data) and contents list. They should be accompanied by the standard text for the inside front cover and, if appropriate, a list of running heads. Information for the front-cover design, text for the back cover and spine, and a list of related WHO publications for the inside back cover should be prepared for GRA (nonserial publications) or sent directly with the main manuscript to the typesetter (series publications).

An English/French list of terms used for manuscript mark-up and proof correction is given in Annex 4.

Tables

Tables should be kept as simple as possible, with brief titles and column headings. Restrict the use of rules, unless the table is particularly complex. Vertical lines should generally be avoided, but straddle rules over column headings are useful for clarifying hierarchical arrangements. When units of measurement or multiplication factors appear in column headings, ensure that there is no possibility of misinterpretation; "Concentration (mg/litre)" is clear, but "No. of cases detected ($\times 10^3$)" is ambiguous. A column should not contain any blanks; use dashes or abbreviations where data are missing (for example NA, ND, for "not applicable", "not determined").

Additional information should be given in the form of footnotes, generally identified with superscript letters placed in the appropriate order in the table from left to right and top to bottom. The same mark may be used on two or more elements if the corresponding note applies to them, for example:

^aData based on a teratogenicity study in which ochratoxin A was administered on day 9 of gestation.

For a table consisting only of words or letters, superscript numbers may be used as reference marks to identify footnotes, and for a table that includes mathematical or chemical equations a series of symbols may be used, because of the risk of mistaking letters or numbers for exponents:

* (asterisk), † (dagger), ‡ (double dagger), § (section mark).

When more symbols are needed, these may be doubled and tripled in the same sequence. Asterisks are conventionally also used to indicate levels of statistical significance.

The footnotes should be printed immediately below the table. They are of four general kinds and should appear in this order:

1. Explanations of abbreviations and notes on levels of statistical significance. A single asterisk is used for the lowest level of significance, two for the next level, and so on, with the specific level given in the following form:

* $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$.

2. Source and general notes. If data for a table are not the author's own but are taken from another source, the author should include a source note, introduced by the word "Source(s):", for example:

Source: Ballance & Lauer (12).

3. Notes on specific parts of the table.
4. Acknowledgements. If the entire table is reproduced with little or no change from another source, this should be acknowledged in a statement such as:

Reprinted, by permission of the publisher, from Crompton DWT et al., eds. *Ascariasis and its public health significance*. London, Taylor & Francis, 1985.

When editing tables, ensure that each one is internally consistent (check, for example, for errors in adding up columns of figures or calculating percentages). Every table must also agree with, and be referred to in, the body of the text. For typesetting, make sure that column alignments, rules and typeface requirements are clearly marked and that footnotes cannot be confused with running text.

Some of these points are illustrated in the example below; the precise format for the title and use of boldface and italic will depend on the style adopted for the publication in question.

Table 1
Concise descriptive title^a

Column heading ^c	Main column heading ^b	
	Extremely long subheading (unit) ^d	Subheading (unit)
Row heading indented on second and subsequent lines	10.3 ± 1.1 ^e	ND
Second row heading	9.1 ± 3.2	362
Subheading	2.3 ± 0.2 [*]	
Subheading	6.8 ± 2.5	
Third row heading	3.0 ± 0.9	490
Final row heading	17.6 ± 4.4 ^{**}	51

ND, not determined; ^{*}*P* < 0.05; ^{**}*P* < 0.01 (with appropriate details of the statistical test).

^a Explanatory note, for example concerning experimental design and method, or source of data.

^b Column headings may be ranged left or centred over the columns as appropriate (usually centred when the columns contain figures).

^c First column heading normally ranged left and aligned with the bottom line of column headings.

^d Explanatory note, for example concerning presentation of data (mean ± SD, etc.).

^e Align columns of figures on the decimal point.

Adapted, by permission of the publisher, from Bloggs & Grundy (16).

Lists

In an effort to rationalize the presentation of lists, the following are the styles that should generally be adopted. Where circumstances make any of these inappropriate, clarity and common sense should prevail in selecting an alternative.

Format 1

1. Where subsequent reference makes it essential that items in a list be numbered, and always for recommendations within a Technical Report, numbers should be set full out to the left and followed by a full point.
2. Items of this sort will often contain more than one sentence. They should start with a capital letter and end with a full point.
3. Where more than one line of text is involved, second and subsequent lines should be blocked to the first tab stop.

Format 2

1. All the rules set out in Format 1 apply in this Format, which deals with items requiring a further subdivision.
2. When an individual item in a list is to be broken down beyond the numbering level, subitems should be identified and set out as follows:
 - (a) by the use of lower-case letters, in parentheses, set at the first tab stop;
 - (b) for long lists, all letters may be used, i.e. including i, l and o;
 - (c) where a subitem runs to more than one line of text, second and subsequent lines should be blocked to the second tab stop;
 - (d) in general, the nature of these subitems is likely to be such that each can be terminated with a semicolon (except the last, which requires a full point).

Format 3

- Where certain parts of a text are to be itemized, for example for emphasis, but there is no requirement for numbering, the use of "bullets" is recommended.
- Bullets should be set full out to the left, and should be of a size appropriate to the appearance of the text.

- In general, such items are likely to consist of one or more complete sentences, and should therefore each begin with a capital letter and end with a full point.
- As in the case of numbered items, where more than one line of text is involved, second and subsequent lines should be blocked to the first tab stop.

Format 4

- The style of a particular text may demand that, for lists of this type, the bullets be set at the first tab stop.
- In such a case, the second and subsequent lines of text should be blocked to the second tab stop.

Format 5

Lists of short items, usually – but not invariably – of one line or less, may often follow a brief introductory statement and should then be set out as follows:

- dash at left-hand margin
- lower-case letter to start
- the second line of longer items blocked to the first tab stop
- no punctuation after any item but the last.

Format 6

This is almost identical with Format 5, except:

- where the text is rendered clearer by punctuation, each item should end with a comma or a semicolon;
- where some or most of the items run to a second line, the overall appearance may be improved by punctuating at the end of each item.

Format 7

Occasionally, the need for particular emphasis demands the use of bullets in place of the dashes specified in Formats 5 and 6.

11. Abbreviations and symbols

The purpose of abbreviations is to save space in figures and tables or when the same word or phrase occurs many times in an article. All abbreviations should be defined the first time they are used, unless likely to be familiar to readers. A few abbreviations (e.g., i.e., viz., etc.) are so widely used that the complete words are almost never given. However, editors should use their discretion when employing abbreviations, since overuse of less familiar abbreviations can be a hindrance to readers. Care should also be taken to avoid the use of abbreviations that may be confusing.

Symbols, which can be alphabetic, alphanumeric or graphic in form, are used to represent quantities, units, substances (e.g. amino acids, immunoglobulins), chromosomes, mathematical operations, etc. They are often (but not necessarily) abbreviations. For further information on specific symbols, see "Statistics", in section 13, and "Scientific terminology", section 14.

Structure of abbreviations and symbols

Abbreviations and symbols may be formed in any one of the following ways:

1. By omitting the end of a word, which should be replaced by a full stop: Jan., Co.
2. By suppressing the middle of a word: Dr, Mr (a full stop is not usually required).
3. By combining the initial letters of the main words, or components of words, in expressions such as titles of organizations, compound names of chemical substances, or certain medical terms: WHO (World Health Organization), DDT (dichlorodiphenyltrichloroethane), CNS (central nervous system). In abbreviations of this kind, capitals should be used without full stops.
4. By using one or more letters of the word in question, or even letters or signs that do not figure in it, to form a symbol (as in units of measurement and chemical elements): J (joule), Pb (lead), μg (microgram).

Where an abbreviation requiring a full stop falls at the end of a sentence, a second full stop need not be added.

Further reading

Armstrong Lowe D. *A guide to international recommendations on names and symbols for quantities and on units of measurement*. Geneva, World Health Organization, 1975 (Progress in standardization, No. 2).

Baron DN, ed. *Units, symbols, and abbreviations; a guide for biological and medical editors and authors*, 4th ed. London, Royal Society of Medicine Services, 1988.

ISO standards handbook: 2. *Units of measurement*, 2nd ed. Geneva, International Organization for Standardization, 1982.

Quantities, units and symbols, 2nd ed. London, The Royal Society, 1975 (amendment 1981).

The SI for the health professions. Geneva, World Health Organization, 1977 (updated and reprinted 1982).

Abbreviations in common use

The list below gives some common abbreviations with indications of their use.

and	& ¹
as desired (<i>ad libitum</i>)	ad lib
compare (<i>confer</i>)	cf.
continued	cont.
edition	ed. ¹
editor(s)	ed., eds ¹
<i>et alii</i>	et al. ¹
<i>et cetera</i>	etc.
<i>et sequentia</i>	et seq.
figure(s)	Fig., Figs ²
for example (<i>exempli gratia</i>)	e.g.
manuscript(s)	MS, MSS
<i>nota bene</i>	N.B.
number(s)	no., No., Nos
page(s)	p., pp.
<i>per centum</i>	per cent, %
revised, revision	rev.
that is (<i>id est</i>)	i.e.
Volume(s)	Vol. ¹

¹ To be used in bibliographic references (see section 9).

² To be used (when appropriate to the style of the book) for a figure, and in cross-references in the text to numbered figures.

Names of organizations

The following abbreviations of international and other organizations are those most frequently encountered in WHO documents and publications.

ACC	Administrative Committee on Coordination
ACHR	Advisory Committee on Health Research
AGFUND	Arab Gulf Programme for United Nations Development Organizations
ASEAN	Association of South-East Asian Nations
CEC	Commission of the European Communities
CIDA	Canadian International Development Agency
CIOMS	Council for International Organizations of Medical Sciences
DANIDA	Danish International Development Agency
EC	European Commission, European Community
ECA	Economic Commission for Africa
ECE	Economic Commission for Europe
ECLAC	Economic Commission for Latin America and the Caribbean
EEC	European Economic Community
ESCAP	Economic and Social Commission for Asia and the Pacific
ESCWA	Economic and Social Commission for Western Asia
EU	European Union
FAO	Food and Agriculture Organization of the United Nations
FINNIDA	Finnish International Development Agency
GATT	General Agreement on Tariffs and Trade
IAEA	International Atomic Energy Agency
IARC	International Agency for Research on Cancer
IBRD	International Bank for Reconstruction and Development (World Bank)
ICAO	International Civil Aviation Organization
ICDO	International Civil Defence Organization
ICMMP	International Committee of Military Medicine and Pharmacy
ICRC	International Committee of the Red Cross
IDA	International Development Association (World Bank)
IFAD	International Fund for Agricultural Development
ILO	International Labour Organisation (Office)
IMF	International Monetary Fund (The Fund)
IMO	International Maritime Organization
IOM	International Organization for Migration
ITO	International Trade Organization

ITU	International Telecommunication Union
NORAD	Norwegian Agency for International Development
OAS	Organization of American States
OAU	Organization of African Unity
OECD	Organisation for Economic Co-operation and Development
OIE	International Office of Epizootics
PAHO	Pan American Health Organization
PASB	Pan American Sanitary Bureau
SAREC	Swedish Agency for Research Cooperation with Developing Countries
Sida	Swedish International Development Cooperation Agency
UICC	International Union against Cancer
UNAIDS	Joint United Nations Programme on HIV/AIDS
UNCTAD	United Nations Conference on Trade and Development
UNDCP	United Nations International Drug Control Programme
UNDP	United Nations Development Programme
UNDRO	Office of the United Nations Disaster Relief Coordinator
UNEP	United Nations Environment Programme
UNESCO	United Nations Educational, Scientific and Cultural Organization
UNFDAC	United Nations Fund for Drug Abuse Control
UNFPA	United Nations Population Fund
UNHCR	Office of the United Nations High Commissioner for Refugees
UNICEF	United Nations Children's Fund
UNIDO	United Nations Industrial Development Organization
UNITAR	United Nations Institute for Training and Research
UNRISD	United Nations Research Institute for Social Development
UNRWA	United Nations Relief and Works Agency for Palestine Refugees in the Near East
UNSCEAR	United Nations Scientific Committee on the Effects of Atomic Radiation
UPU	Universal Postal Union
USAID	United States Agency for International Development
WCC	World Council of Churches
WFP	World Food Programme
WFUNA	World Federation of United Nations Associations
WIPO	World Intellectual Property Organization
WMO	World Meteorological Organization
WTO	World Trade Organization

Units of measurement

Detailed lists of the symbols to be used for units of measurement can be found in the reference sources listed on page 38. A list of the most commonly used symbols (including those used for the SI base units) is given below. These symbols should only be used after a quantity expressed in figures, in tables, and in graphs.

ampere	A	metre	m
becquerel	Bq	minute (of arc)	'
candela	cd	minute (of time)	min
coulomb	C	mole	mol
decibel	dB	newton	N
degree (angular)	°	ohm	Ω
degree Celsius	°C	pascal	Pa
degree Fahrenheit	°F	radian	rad
farad	F	röntgen	R
gram	g	second (of arc)	"
gray	Gy	second (of time)	s
henry	H	sievert	Sv
hour	h	tesla	T
hertz	Hz	tonne	t
joule	J	volt	V
kelvin	K	watt	W
kilogram	kg	weber	Wb
litre	l*		

* Use of this abbreviation should be avoided if confusion is possible.

A compound abbreviation may sometimes be used to indicate a relationship between two different units of measurement: mg/kg.

The SI incorporates the following prefixes, by means of which it is possible to form multiples of SI units. (For guidance on the use of SI units, see section 12.)

Factor	Prefix	Symbol	Factor	Prefix	Symbol
10 ⁻¹	deci	d	10 ¹	deca	da
10 ⁻²	centi	c	10 ²	hecto	h
10 ⁻³	milli	m	10 ³	kilo	k
10 ⁻⁶	micro	μ	10 ⁶	mega	M
10 ⁻⁹	nano	n	10 ⁹	giga	G
10 ⁻¹²	pico	p	10 ¹²	tera	T
10 ⁻¹⁵	femto	f	10 ¹⁵	peta	P
10 ⁻¹⁸	atto	a	10 ¹⁸	exa	E

Abbreviations used in the medical sciences

Extensive lists of abbreviations used in the medical sciences are given in the following publications:

Heister R. *Dictionary of abbreviations in medical sciences*. Berlin, Springer Verlag, 1989.

Steen EB. *Baillière's abbreviations in medicine*, 5th ed. London, Baillière Tindall, 1984.

The following acronyms and abbreviations are in common use in the medical sciences; those marked with an asterisk do not usually need to be defined.

ACTH	adrenocorticotrophic hormone
ADP*	adenosine diphosphate
AIDS	acquired immunodeficiency syndrome
AMP*	adenosine monophosphate
ATP*	adenosine triphosphate
BCG*	bacille Calmette–Guérin (vaccine)
BMR	basal metabolic rate
BOD	biochemical oxygen demand
BP	blood pressure
CAAT	computer-assisted axial tomography
cDNA	complementary deoxyribonucleic acid
CIEP	counterimmunoelectrophoresis
CMV	cytomegalovirus
CNS	central nervous system
CSF	cerebrospinal fluid
DNA*	deoxyribonucleic acid
DTP	diphtheria–tetanus–pertussis vaccine
EBV	Epstein–Barr virus
ECG	electrocardiogram, electrocardiography
ECT	electroconvulsive therapy
ED₅₀	median effective dose
EEG	electroencephalogram, electroencephalography
ELICT	enzyme-linked immunocytochemical technique
ELIEDA	enzyme-linked immunoelectron diffusion assay
ELISA	enzyme-linked immunosorbent assay
EMIT	enzyme multiplication immunoassay technique
F₁*	first filial generation
FAST	fluorescent antibody staining technique; fluoro-allergosorbent test
FAT	fluorescent antibody technique (test)

FSH	follicle-stimulating hormone
GABA	gamma-aminobutyric acid
GH	growth hormone
GnRH	gonadotropin-releasing hormone
HAT	haemagglutination antibody test
HBV	hepatitis B vaccine (virus)
HCG	human chorionic gonadotropin
HIV	human immunodeficiency virus
HLA	histocompatibility leukocyte (locus) antigen; human leukocyte (lymphocyte) antigen
HPLC	high-pressure (performance) liquid chromatography
HTLV	human T-cell lymphotropic virus
MHC	major histocompatibility complex
IDT	immunodiffusion test
IFAT	indirect fluorescent antibody test
IFT	immunofluorescence test
Ig	immunoglobulin (IgA*, IgD*, IgE*, IgG*, IgM*)
IR	infrared
LAT	latex agglutination test
LD₅₀	median lethal dose
LFT	latex flocculation test
LH	luteinizing hormone
MAb	monoclonal antibody
mRNA	messenger ribonucleic acid
NSAID	non-steroidal anti-inflammatory drug
PAGE	polyacrylamide gel electrophoresis
PCV	packed cell volume
PMSG	pregnant mare serum gonadotropin
RAST	radioallergosorbent test
RFLP	restriction fragment length polymorphism
RISA	radioimmunosorbent assay
RIST	radioimmunosorbent test
RNA*	ribonucleic acid
rRNA	ribosomal ribonucleic acid
SDS	sodium dodecyl sulfate
SDS-PAGE	sodium dodecyl sulfate-polyacrylamide gel electrophoresis
sp., spp.*	species
TCID₅₀	median tissue culture infective dose
TLC	thin-layer chromatography
tRNA	transfer ribonucleic acid
UV	ultraviolet

Units of currency

Lists of the official basic units of currency and the appropriate currency abbreviations are given in:

Currency units. New York, United Nations, 1993 (Terminology bulletin no. 346).

The currencies listed are those officially in use as at 19 March 1993 and may be subject to change. Further information can be obtained from CTT. The abbreviations used for the various currencies are also given in the following publication:

Hunter B, ed. *The Statesman's year-book: statistical, political and economic account of the states of the world for the year 1997–1998*. New York, St Martin's Press, 1997.

When two or more countries use the same name for their currencies, the appropriate distinguishing adjective should be used at first mention, or throughout the text if confusion is possible: two million Irish pounds, six million pounds sterling.

When the monetary unit is written in full, it should follow the number for the amount, as in the examples above, but when the abbreviation is used, it should always precede the amount. Where a symbol is used to represent the monetary unit, there should be no space between the symbol and the amount, for example:

£19.95.

Where the abbreviation is a letter or letters or a combination of letter(s) and symbol, there should be one space between the abbreviation and the amount, for example:

US\$ 6000.

WHO's three-letter currency codes (GBP, USD, CHF, etc.) are for internal use only, and should not appear in publications.

Compass points

Compass points should be abbreviated in indications of latitude and longitude, and in all tables and figures. Capital letters should be placed immediately after each other without the interposition of full stops or spaces: from latitude 7°10'N to 0°40'S, SSE.

States, territories and districts of the United States of America

The following two-letter abbreviations may be used in addresses, reference lists, figures and tables.

Alabama	AL	Kentucky	KY	Ohio	OH
Alaska	AK	Louisiana	LA	Oklahoma	OK
Arizona	AZ	Maine	ME	Oregon	OR
Arkansas	AR	Maryland	MD	Pennsylvania	PA
California	CA	Massachusetts	MA	Puerto Rico	PR
Canal Zone	CZ	Michigan	MI	Rhode Island	RI
Colorado	CO	Minnesota	MN	South Carolina	SC
Connecticut	CT	Mississippi	MS	South Dakota	SD
Delaware	DE	Missouri	MO	Tennessee	TN
District of Columbia	DC	Montana	MT	Texas	TX
Florida	FL	Nebraska	NE	Utah	UT
Georgia	GA	Nevada	NV	Vermont	VT
Hawaii	HI	New Hampshire	NH	Virginia	VA
Idaho	ID	New Jersey	NJ	Washington (State)	WA
Illinois	IL	New Mexico	NM	West Virginia	WV
Indiana	IN	New York	NY	Wisconsin	WI
Iowa	IA	North Carolina	NC	Wyoming	WY
Kansas	KS	North Dakota	ND		

12. ***Numbers, dates, time and units of measurement***

Numbers

In general, words should be used for the numbers one to nine and at the beginning of sentences, and figures for 10 and above; however, editors should use their judgement in individual cases. Figures should nevertheless always be used in specific numerical contexts (e.g. “day 3” or “a factor of 2”). Numbers that constitute a series or range should usually be written as figures if at least one of them is greater than nine, for example “we treated 19 patients with drug A and 7 with drug B”.

When a number has more than four digits, it should be split by means of a thin space (not a comma) after every three digits to the left or right of the decimal point (e.g. 10 000, 0.234 56), but in a table column containing a mixture of numerals with four digits and numerals with more than four digits, align them by spacing all of them as for the latter group.

A decimal point should always be preceded by a figure, if necessary a zero. Very small numbers are usually best given in exponential notation, for example 1.6789×10^{-6} .

The term “billion” is ambiguous (meaning 10^9 in American and often 10^{12} in British usage) and should be defined or avoided.

When a colon is used to indicate a ratio, there should be a thin space on either side of it.

Dates

Dates should be written in the form 12 September 1989.

Time

For the time of day, the 24-hour clock should generally be used, e.g. 16:30.

Units of measurement

Measurements should normally be given in SI units in accordance with *The SI for the health professions*; for summary tables, see section 11. Most SI symbols are lower-case letters except those derived from proper names (e.g. Pa, Gy, W). Symbols are identical in the singular and

plural, and are not followed by a full stop unless they occur at the end of a sentence.

Figures should always be used with unit symbols and abbreviations, for example 7 kg not seven kg; the figure should be separated from the unit symbol by a thin space. To avoid ambiguity, no more than one solidus should be used to divide units, for example m/s^2 (or $\text{m}\cdot\text{s}^{-2}$) not m/s/s . When a medial multiplication point is used, it should be clearly identified.

Exceptionally, for blood pressure, values may still be given in millimetres of mercury with the equivalent in kilopascals in parentheses, for example

120 mmHg (16 kPa),

and in texts on nutrition, values for energy may be quoted in both joules and thermochemical calories, for example

5.69 MJ (1360 kcal_{th}).

Concentration

The term "concentration" has been applied to numerous different quantities. The appropriate units for measuring three of the most common are: substance concentration, mol/m^3 (or mol/l); mass concentration, kg/m^3 (or kg/l); and mass fraction, kg/kg . Wherever possible, the first of these should be used. The symbol *M* should not be substituted for mol/l , and the term "molarity" should not be used.

Relative molecular mass

The relative molecular mass of a substance (M_r) is the ratio of the mass of one molecule of that substance to one-twelfth of the mass of an atom of carbon-12. It is therefore a pure number and dimensionless. The term "molecular weight" should no longer be used.

Temperature

Temperature should be given in the form 7 °C or 20–25 °C. The degree sign is part of the unit and should be closed up to the C.

Further reading

Armstrong Lowe D. *A guide to international recommendations on names and symbols for quantities and on units of measurement*. Geneva, World Health Organization, 1975 (Progress in standardization, No. 2).

The SI for the health professions. Geneva, World Health Organization, 1977 (updated and reprinted, 1982).

13. *Mathematics and statistics*

Mathematical matter and equations

Italics should be used for mathematical variables and certain physical constants, roman type for operators (such as +, =, d and multiplication and integral signs), abbreviations (such as log, sin and exp) and representations of pure numbers (such as e and i).

Badly written or ambiguous symbols and signs should be identified for the typesetter (e.g. italic letter “x”/Greek chi/multiplication sign; capital letter “O”/Greek theta/zero), and the spatial relationship of terms made clear. When several brackets have to be used in a mathematical expression, the sequence should be { [(...)] }.

Mathematical formulae should be written in such a way as to take up as little space as possible, for example

$$x/(a-b) \text{ not } \frac{x}{(a-b)},$$

except when this could cause difficulty in understanding or lead to confusion.

Mark thin spaces before and after mathematical operators, except when + or – is used to indicate a positive or negative number (e.g. –6) and around a medial multiplication point or solidus.

Equations for display should be clearly marked and guidance provided on where to break those that are likely to be too long to fit on one line (preferably before an operational sign and not in the middle of a term). The second line may be set flush right, a standard indentation from the right margin, or aligned on operational signs. Equations (whether in line or displayed) that form part of sentences may need punctuation depending on their context.

Statistics

Symbols recommended by the International Organization for Standardization are shown in the table overleaf.

Term	Symbol
variable in a population	$X, Y, \dots$
particular or observed value	$x, y, \dots$
population or lot size	N
sample size	n
range of a sample	w, R
arithmetic mean of a population	$\bar{X}$
arithmetic mean of a sample	$\bar{x}$
variance of a random variable or of a population	σ^2
variance of a sample	s^2
standard deviation of a random variable or of a population	σ
standard deviation of a sample	s
coefficient of correlation between two random variables or in a population	ρ
coefficient of correlation in a sample	r
number of degrees of freedom	v
standard normal variable	U, Z
particular value of the standard normal variable	u, z
chi-squared distribution	χ^2
t-distribution (Student)	t
F-distribution	F
level of significance of a test, type I risk	α
type II risk	β
probability of an event E	$P(E)$

Several abbreviations are in common use in running text, for example:

- CI confidence interval
- df degrees of freedom
- SD standard deviation
- SE standard error.

In statistical analyses, the mean should be supported by statistics such as the number of observations, the standard deviation (a measure of the variability of a set of data) or the standard error (a measure of the precision of an estimate, commonly the mean). Avoid presenting variability in the form $SE = \pm 1.3$, which wrongly implies that SE can be negative. Probability values are usually quoted in the form " $P < 0.001$ ".

Further reading

Altman DG et al. Statistical guidelines for contributors to medical journals. *British medical journal*, 1983, 286:1489–1493.

Gardner MJ, Altman DG. Confidence intervals rather than P values: estimation rather than hypothesis testing. *British medical journal*, 1986, 292:746–750.

Swinscow TDV. *Statistics at square one*, 9th ed. (revised). London, BMJ Publishing Group, 1996.

14. **Scientific terminology**

The scientific terminology used in WHO publications should be in accordance with the recommendations of the relevant international bodies. Guidance on correct usage can be obtained from CTT.

Anatomy

In general, the anglicized versions of Latin anatomical terms should be used, as found in the standard medical dictionaries. If, for a specific reason, Latin terms are preferred in a given context, they should not be italicized.

Animals, plants, bacteria and viruses

The relevant codes of international nomenclature should be used.

The name of a genus is set in italic with an initial capital (e.g. *Staphylococcus*); it refers to all species and subspecies within the genus and should not be used as a substitute for the name of a single species. The name of a species consists of the genus name and the specific epithet, both italicized (e.g. *Escherichia coli*). The name of the person who originally classified the species sometimes follows the binomen in roman (e.g. *Schistosoma mansoni* Sambon). A genus name may be abbreviated to its initial letter after the first mention, unless this leads to ambiguity.

Common names (not italicized) also exist for certain genera and species, for example:

pseudomonad
salmonella
blackfly
gorilla.

Both one-word and two-word common names are in use for viruses (e.g. poliovirus, papillomavirus, poxvirus, herpesvirus; but influenza virus, cowpox virus, rubella virus, mumps virus).

It should be noted that the bacterial and viral codes of nomenclature differ substantially; CTT should be consulted for details. A full list of approved bacterial names that have been published in the *International journal of systematic bacteriology* is available under WHOTERM through the WHO Web site.

Further reading

International code of zoological nomenclature adopted by the 20th General Assembly of the International Union of Biological Sciences, 3rd ed. London, International Trust for Zoological Nomenclature, 1985.

Lapage SP, Sneath PH, eds. *International code of nomenclature of bacteria*. Washington, DC, American Society for Microbiology, 1992.

Murphy FA et al., eds. *Virus taxonomy. Sixth report of the International Committee on Taxonomy of Viruses*. Vienna, Springer, 1995 (Archives of virology, suppl. 10).

Skerman VBD, McGowan V, Sneath PHA, eds. *Approved lists of bacterial names*, amended ed. Washington, DC, American Society for Microbiology, 1989. [See also additional lists published in the *International journal of systematic bacteriology*.]

Biochemistry

The recommendations of the IUPAC–IUB Joint Commission on Biochemical Nomenclature and of the Nomenclature Committee of IUB¹ should be followed, details of many of which are to be found in Liébecq C. *Biochemical nomenclature and related documents, a compendium*, 2nd ed. (London, Portland Press, 1992).

Amino acids

The three-letter symbols in the table opposite may be used for representing polymers or sequences and in figures and tables; a known sequence of amino acids is indicated by hyphens between the symbols. The use of one-letter symbols should be restricted to the comparison of long sequences.

¹ Recently renamed the International Union of Biochemistry and Molecular Biology (IUBMB).

Common name	3-letter symbol	1-letter symbol
alanine	Ala	A
arginine	Arg	R
asparagine	Asn	N
aspartic acid	Asp	D
cysteine	Cys	C
glutamine	Gln	Q
glutamic acid	Glu	E
glycine	Gly	G
histidine	His	H
isoleucine	Ile	I
leucine	Leu	L
lysine	Lys	K
methionine	Met	M
phenylalanine	Phe	F
proline	Pro	P
serine	Ser	S
threonine	Thr	T
tryptophan	Trp	W
tyrosine	Tyr	Y
valine	Val	V
unspecified amino acid	Xaa	X

Enzymes

The recommendations (names and numbers) of the latest edition of *Enzyme nomenclature* and its supplements should be followed as far as possible. IUB recommendations also exist for symbolism and terminology in enzyme kinetics.

Further reading

Enzyme nomenclature, 1992: recommendations of the Nomenclature Committee of the International Union of Biochemistry and Molecular Biology on the nomenclature and classification of enzymes. San Diego, Academic Press, 1992.

Nomenclature Committee of the International Union of Biochemistry. Symbolism and terminology in enzyme kinetics. *European journal of biochemistry*, 1982, 128:281–291.

Nucleosides

The symbols used for ribonucleoside residues (radicals) are:

A	adenosine	C	cytidine
G	guanosine	T	ribosylthymine
I	inosine	U	uridine
X	xanthosine	ψ	5-ribosyluracil (pseudouridine).

The prefix "r" (for ribo) may be used for emphasis or clarity.

The 2'-deoxyribonucleosides can be designated by the same symbols preceded by "d", for example:

dA 2'-deoxyribosyladenine
dT 2'-deoxyribosylthymine (thymidine).

Sugars

Symbols may be used for representing polymers or sequences and in figures and tables:

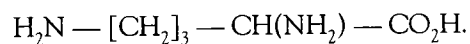
Ara	arabinose	Man	mannose
dRib	2-deoxyribose	Rib	ribose
Fru	fructose	Neu	neuraminic acid
Fuc	fucose	Rha	rhamnose
Gal	galactose	Xyl	xylose.
Glc	glucose		

Chemistry

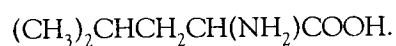
Chemical names should follow the IUPAC rules, as interpreted by the American Chemical Society (see *Chemical abstracts*). Note that "sulfur" should always be used rather than "sulphur", but that the British spellings "aluminium" and "caesium" are preferred to "aluminum" and "cesium".

General

Chemical names should be spelt out in text unless the symbolic formula is graphically useful. In one-line formulae used to represent molecules, dashes may be used to represent bonds in the main chain, for example:



If there is no need to show linkages, the formula may be written without bond symbols, for example:



"C₂₀ acid" is used to denote an acid containing 20 carbon atoms and "C-3" to denote the carbon atom numbered 3.

Oxidation numbers are indicated by **roman numerals** in parentheses closed up against the name of the element concerned, for example:

manganese(IV)
Cr(III).

Further reading

Budavari S et al., eds. *The Merck index*, 12th ed. Whitehouse Station, NJ, Merck, 1996.

Cahn RS, Dermer OC. *Introduction to chemical nomenclature*, 5th ed. London, Butterworths, 1979.

IUPAC Commission on the Nomenclature of Inorganic Chemistry. *Nomenclature of inorganic chemistry*, 3rd ed. Oxford, Blackwell Scientific, 1990.

IUPAC Commission on the Nomenclature of Organic Chemistry. *Nomenclature of organic chemistry: sections A, B, C, D, E, F and H*, revised ed. Oxford, Pergamon, 1979.

Isotopic labelling

The symbol for the isotope introduced is placed in square brackets directly attached to the part of the name to which it refers (e.g. sodium [¹⁴C]formate); appropriate arabic numerals and prefixes are added as necessary (e.g. L-[2-¹⁴C]leucine). For simple molecules, chemical formulae may be used (e.g. ¹⁴CO₂).

Square brackets should not be used when the isotopic symbol is attached to a word that is not a chemical name (e.g. "¹⁴C-labelled"). Terms such as "¹³¹I-labelled albumin" should not be contracted to [¹³¹I]albumin since native albumin does not contain iodine.

In text, an isotope can be identified in the following form:

phosphorus-32 iodine-131.

Prefixes

Various stereochemical prefixes are in use, including:

(R)-	cis-	D-	(E)-	(+)-	(±)-
(S)-	trans-	L-	(Z)-	(-)-	meso-

The older prefixes *dextro*, abbreviated *d* and equivalent to (+), and *laevo*, abbreviated *l* and equivalent to (-), should no longer be used.

Certain prefixes should be italicized (see also above):

o-, *m*-, *p*- (but these locants have generally been abandoned by
Chemical abstracts in favour of numerals)

N-, O-, S- (to indicate an element carrying a substituent).

Italics are not, however, used for:

allo	bis	cyclo	epi
iso	nor	tris.	

A lower-case italic prefix at the beginning of a line or after a full stop is not given an initial capital; the next letter becomes a capital instead (e.g. *trans*-2-Butene). A capitalized italic prefix does not serve as the capital marking the beginning of a sentence; the first letter of the name itself should therefore become a capital (e.g. *N,N*-Dimethyltryptamine). Italicized syllables are not “counted” in alphabetization.

Diseases

As far as possible the International Nomenclature of Diseases should be followed for disease names and their spelling; otherwise standard dictionaries (e.g. Stedman's or Dorland's) should be consulted. It should be noted that the International Classification of Diseases (ICD-10) is a statistical classification and is not intended to be a reference for nomenclature.

Further reading

Critchley M, ed. *Butterworths medical dictionary*, 2nd ed. London, Butterworths, 1978.

Dorland's illustrated medical dictionary, 28th ed. Philadelphia, Saunders, 1994.

International nomenclature of diseases [several volumes published by CIOMS or CIOMS/WHO].

International Statistical Classification of Diseases and Related Health Problems, tenth revision. Volume 1: tabular list; Volume 2: instruction manual; Volume 3: alphabetical index. Geneva, World Health Organization, 1992–1994.

Landau SI, ed. *International dictionary of medicine and biology*. New York, Wiley, 1986.

Stedman's medical dictionary, 26th ed. Baltimore, Williams & Wilkins, 1995.

Drugs and pesticides

For pharmaceuticals the International Nonproprietary Names (INNs) should be used, and for pesticides the common names adopted by the International Organization for Standardization (see International Standard ISO 1750:1981, and subsequent addenda and amendments). When no internationally approved name exists, an approved national name may be used. Proprietary names should be avoided whenever possible.

Further reading

International nonproprietary names (INN) for pharmaceutical substances, cumulative list no. 9. Geneva, World Health Organization, 1996.

IPCS, International Programme on Chemical Safety. *The WHO recommended classification of pesticides by hazard and guidelines to classification 1996–1997*. Geneva, World Health Organization, 1996 (unpublished document WHO/PCS/96.3) [contains an index of pesticides listed by their common names].

Copping LG, ed. *Pesticide index: an index of chemical, common and trade names of pesticides and related crop-protection products*. Farnham, British Crop Protection Council, 1995.

Marler EEJ. *Pharmacological and chemical synonyms*, 10th ed. Amsterdam, Elsevier, 1994.

Tomlin C, ed. *The pesticide manual: a world compendium*, 10th ed. Farnham, British Crop Protection Council, 1994.

Genetics**Genes and chromosomes**

The International Committee on Standardizing Human Gene Nomenclature has developed a system known as the International System for Human Gene Nomenclature, but this is not universally accepted. Human genes are generally denoted by italicized symbols composed of capital letters, sometimes with the addition of an arabic numeral (e.g. CF, G6PD). Lower-case letters are commonly still used in blood-group symbols (e.g. Fy, Rh). Phenotype symbols are set in roman type (e.g. ADA 1).

Human chromosomes may be referred to by number or letter (1–22, X, Y) or by group (A–G).

For bacteria, genotype symbols usually consist of three italic lower-case letters (e.g. *ara*) and phenotype symbols of three roman letters, the first of which is a capital (e.g. Met).

Further reading

Shows TB et al. International system for human gene nomenclature (1981) – ISGN 1981. *Cytogenetics and cell genetics*, 1982, 32:221–245.

Shows TB et al. Guidelines for human gene nomenclature: an international system for human gene nomenclature (ISGN 1987). *Cytogenetics and cell genetics*, 1987, 46:12–28.

Oncogenes

Oncogenes are usually designated by three-letter italic terms, often taken from the names of the viruses with which the genes are

associated. A roman prefix may be added: "v-" for viral, "N-" for nuclear and "c-" for the gene's cellular counterpart (proto-oncogene). Other prefixes and various suffixes are common. For example:

Ha-ras-1	N-ras	v-erb B
yes	c-abl	fes/fps.

Restriction enzymes

The generally used nomenclature for restriction enzymes follows, more or less, the recommendations of Smith & Nathans (1973). The genus and species name of the host organism is identified by the first letter of the genus and the first two letters of the species to form a three-letter abbreviation in italic (e.g. *Escherichia coli* becomes *Eco*). The strain or type identification (or symbol for an extrachromosomal element) follows in roman type (e.g. *Eco*P1, *Hind*). Roman numerals are used to identify different restriction-modification systems; they are separated from the enzyme designation by a space only if there is no strain designated. For example:

<i>Bam</i> HI	<i>Eco</i> RI	<i>Hind</i> III
<i>Xma</i> I	<i>Hae</i> II.	

Further reading

Smith HO, Nathans D. A suggested nomenclature for bacterial host modification and restriction systems and their enzymes. *Journal of molecular biology*, 1973, 81:419-423.

Haematology

Blood groups

Blood groups (A, B, O, Le, Fy, Rh etc.) are set in roman, as are the antigens after which the groups are named. The genes corresponding to the antigens are usually denoted by italic letters (e.g. *A*).

Further reading

Issitt PD, Crookston MC. Blood group terminology: current conventions. *Transfusion*, 1984, 24:2-7.

Clotting factors

Clotting factors are usually designated by roman numerals (e.g. factor IX). An "a" added to the numeral indicates the activated form of the factor (e.g. factor VIIa).

Haemoglobins

The term “haemoglobin” is usually represented by the symbol Hb and the various types of haemoglobin by capital letters or the names of the laboratories, hospitals or places in which they were found (e.g. Hb F, Hb S, Hb Hammersmith). The major and minor adult haemoglobins are designated Hb A and Hb A₂. The four polypeptide chains of normal human haemoglobin are designated by the Greek letters α , β , γ , and δ and the paired chains in the form $\alpha_2\beta_2$, $\alpha_2\delta_2$ etc. For abnormal haemoglobins, the locations of amino acid differences may be indicated by designating the chain, the numbered position on it, the helix position in parentheses and the amino acid substitution, for example:

$\alpha 87(\text{F8})\text{His} \rightarrow \text{Tyr}$ for Hb M Iwate.

Further reading

International Society of Haematology. Nomenclature of abnormal haemoglobins. [Recommendations unanimously accepted at the Tenth International Congress of Haematology, Stockholm, 1964.] *British journal of haematology*, 1965, 11:121–122.

Tenth International Congress of Haematology. Nomenclature of haemoglobins. *British medical journal*, 1964, 2:1258.

Immunology**Immunoglobulins**

Immunoglobulins should be represented by the symbol “Ig” and a capital letter designating the specific class:

IgG IgA IgM IgD IgE.

Subclasses are indicated by a number attached to the class symbol, for example:

IgG2 IgM3.

Further reading

An extension of the nomenclature for immunoglobulins. *Bulletin of the World Health Organization*, 1969, 41:975–978.

Nomenclature for human immunoglobulins. *Bulletin of the World Health Organization*, 1964, 30:447–450.

Nomenclature of human immunoglobulins. *Bulletin of the World Health Organization*, 1973, 48:373.

Notation for genetic factors of human immunoglobulins. *Bulletin of the World Health Organization*, 1965, 33:721–724.

Notation for human immunoglobulin subclasses. *Bulletin of the World Health Organization*, 1966, 35:953.

Rowe DS. Nomenclature of immunoglobulins. *Nature*, 1970, 228:509–511.

Interferons

Interferons are represented by the symbol IFN to which is attached by a hyphen the Greek letter representing its type, for example IFN- α for alpha interferons. Non-allelic variants can be indicated by an arabic number or capital letter (sometimes subscripted) attached to the Greek letter. The prefixes "Hu" and "Mu" may be used to represent "human" and "murine" respectively. For example:

MuIFN- γ HuIFN- β HuIFN- α 4.

Further reading

Interferon nomenclature. *Journal of immunology*, 1980, 125:2353.

Interferon nomenclature. *Journal of general virology*, 1984, 65:669–670.

Interleukins

Designations have been officially adopted for 17 interleukins: interleukin-1 to interleukin-16, including interleukin-1 α and interleukin-1 β . These terms are generally abbreviated IL-1 α , IL-5, IL-13, etc.

Further reading

IUIS/WHO Standing Committee on Interleukin Designation. Nomenclature for secreted regulatory proteins of the immune system (Interleukins): update. *Bulletin of the World Health Organization*, 1997, 75:175.

Miscellaneous

Other recommendations published by WHO

In addition to the books and articles mentioned above, WHO has published terminological recommendations on the following topics (among others):

Allergens

IUIS/WHO Allergen Nomenclature Subcommittee. Allergen nomenclature. *Bulletin of the World Health Organization*, 1994, 72:797–806.

Complement

Nomenclature of complement. *Bulletin of the World Health Organization*, 1968, 39:935–938.

Nomenclature of the alternative activating pathway of complement. *Bulletin of the World Health Organization*, 1981, 59:489–491.

WHO-IUIS Nomenclature Sub-Committee. Nomenclature for human complement component C2. Revised nomenclature for human complement component C4. Nomenclature for human complement factor B. *Bulletin of the World Health Organization*, 1992, 70:527–528, 531–535, 541–543.

HLA system

IUIS-WHO Nomenclature Subcommittee. Nomenclature for clusters of differentiation (CD) defined on human leukocyte populations. *Bulletin of the World Health Organization*, 1984, 62:809–811.

Nomenclature for factors of the HLA system 1977. *Bulletin of the World Health Organization*, 1978, 56:461–465.

Nomenclature for factors of the HLA system 1980. *Bulletin of the World Health Organization*, 1981, 58:945–948.

Nomenclature for factors of the HLA system 1984. *Bulletin of the World Health Organization*, 1985, 63:399–405.

Later recommendations of the WHO Nomenclature Committee concerning the HLA system have been published elsewhere – see, for example, *Immunobiology*, 1988, 177:465–476 and *Immunogenetics*, 1990, 31:131–140.

Vitamins

The familiar names of most vitamins have INN equivalents, for example:

vitamin A	retinol
B ₁	thiamine
B ₂	riboflavin
B ₁₂	cyanocobalamin
C	ascorbic acid
D ₂	ergocalciferol
D ₃	colecalciferol
K ₁	phytomenadione
K ₃	menadione.

15. *Non-discriminatory language*

Of the examples of discrimination that are most likely to be encountered in manuscripts, the most common involve discrimination on the grounds of sex. For this reason, the guidelines presented here concentrate largely on means of avoiding overt or implied sex discrimination. However, examples of racial or cultural discrimination may also be found, and many of the principles that apply to the elimination of sex discrimination can also be applied to the eradication of other types of bias.

Modifying the way people speak and write is not easy, and there is – and will continue to be – considerable resistance to change. Unless the use of non-discriminatory language becomes standard practice, however, the achievement of sexual, racial and cultural equality will be hindered.

The importance of non-discriminatory language

Women's role in society can be unconsciously diminished or trivialized by a failure to mention them specifically in writing – thus implying that they are less important than men.

By excluding women, sexist language can cause offence. It is the mistaken belief of many authors that their use of terms such as “man”, “he” and “him” will be assumed by their readers to include women. Most readers make no such assumptions. Moreover, the use of exclusively masculine nouns and pronouns to denote both men and women can become hopelessly ambiguous and lead to confusion, as in the following example:

“As for man, he is no different from the rest. His back aches, he ruptures easily, his women have difficulties in childbirth.”

Further examples demonstrate how sexist writing reinforces many common but inaccurate concepts of the roles of men and women in society:

“Research scientists often neglect their wives and children.”

“It is a great secret of doctors, known only by their wives . . . that most things get better by themselves.”

Non-sexist language – a cumbersome solution?

With care, the results of rewriting text to eliminate discriminatory language will be no less concise than the original – but will be unambiguous and will avoid causing offence. For example:

“He may see himself as the pivot of all that happens. Hallucinations, especially of hearing, are common and may comment on the patient or address him.”

is no more cumbersome when modified as follows:

“Patients may see themselves as the pivot of all that happens. Hallucinations, especially of hearing, are common and may comment on or address patients.”

Similarly:

“Onset of disease and the problem of recognition when the patient starts to observe changes in his skin (e.g. whitish spots, rashes, thickening of skin).

Emerging deformities and the problem of uncertainty: patient starts to feel, and then complain about, the possible effects of the skin changes on his general well-being.”

can be rendered equally well as follows:

“Onset of disease; problems of recognition when patients start to observe skin changes...

Emerging deformities; problems of uncertainty; patients start to feel and then complain about the effects of skin changes on their general well-being.”

In reporting research of any kind that involves both men and women and/or children of both sexes, both males and females should be mentioned in the results. In statistical reports, gross figures should be broken down by sex unless it has previously been established that there is no significant difference between the sexes.

Illustrations and photographs for inclusion in publications should be carefully chosen to ensure that the sexes are equally represented, both in number and in the roles they fulfil. Women, as well as men, should be depicted in professional roles and men, as well as women, in caring roles (such as looking after children).

The following table gives some examples of common discriminatory phrases and suggestions for alternatives.¹

Common discriminatory phrases	Useful solutions	Useful general rules
The client is usually the best judge of the value of his counselling	The client is usually the best judge of the value of counselling The best judge of the value of counselling is the client	Delete redundant possessive pronouns Rephrase
Man's search for knowledge has led him into ways of learning that bear examination	The search for knowledge has led us into ways of learning that bear examination	Use the first person plural
Man, mankind	People, humanity, humans, human beings, humankind	Find alternative words
Man's achievements	Human achievements	
The average man, the man in the street	The average person, people in general	
Man a project	Staff a project, hire personnel, employ staff	
Manpower	Workforce, personnel	
Subjects were 16 boys and 16 girls. Each child was to place a car on his board so that the two boards looked alike	Each child was to place a car on his or her board so that the two boards looked alike	"His or her" is sometimes difficult to avoid and is acceptable if used sparingly
Research scientists often neglect their wives and children	Research scientists often neglect their spouses and children	Use neutral terms such as spouse rather than wife or husband
Woman doctor, male nurse	Doctor, nurse	Specify sex only if necessary to the sense

¹Most of these examples are taken, with permission, from the *Publication manual of the American Psychological Association*, 3rd ed. Washington, DC, American Psychological Association, 1983. This material may be reproduced in whole or in part without permission, provided that acknowledgement is made to the American Psychological Association.

<i>Common discriminatory phrases</i>	<i>Useful solutions</i>	<i>Useful general rules</i>
Foreman, workman, office girl, policeman	Supervisor, worker, secretary, receptionist, office assistant, police officer	Use neutral or equivalent job titles
Show the mother how to cook nutritious meals	Show members of the family how to cook nutritious meals	Do not assume that women always do the household chores
Ask mothers to bring their children to the clinic each month	Ask parents to bring their children to the clinic each month	Remember that men have responsibility for children too
The author acknowledges the assistance of Mrs John Smith	The author acknowledges the assistance of Jane Smith	Use the name the person prefers
Men and women, sons and daughters, boys and girls, husbands and wives	Women and men, daughters and sons, girls and boys, wives and husbands	Vary the order sometimes, unless the traditional order is essential to the sense
Ambitious men, aggressive women; cautious men, timid women	Ambitious people; aggressive people; cautious people; timid people	Avoid implied judgements – just describe people as they are
The boys chose typically male toys The client's behaviour was typically female	The boys chose (specify) The client's behaviour was (specify)	Be specific: it reduces the chances of stereotyping
The client's husband lets her teach part-time	The client teaches part-time	Look out for hidden stereotypes. The intention here was to give the working status of the client; instead it revealed an assumption about husband/wife relationships

The occasional exception

There will be occasions when a term including “man” may be unavoidable, even when the context is not limited to men. An obvious example is “man-made disasters”, for which no generally acceptable alternative is available. Use of the word “Chairman” rather than the more general “Chair” or “Chairperson” may be obligatory. It would be wrong to be over-dogmatic in this area: editors should exercise their judgement, based on the principles outlined here.

Avoiding other types of discrimination

In assessing whether a particular text could cause offence on racial or cultural terms, try substituting your own ethnic or cultural group for that under discussion. If the revised text causes you to feel belittled or offended, you can safely assume that the original was guilty of bias.

Other forms of language that may cause offence include “ageist” terms such as “geriatrics”. Avoid this type of discrimination by substituting less impersonal words such as “older people” or “elderly patients”. Care should also be taken in describing people with disabilities; suggestions for appropriate terminology are given in the Guidelines produced by the Research and Training Center on Independent Living.

People who are ill should be referred to as “patients”: use “diabetic patients” or “patients with hypertension” rather than the dehumanizing “diabetic cases” or “hypertensives”.

Avoid terms such as “the western world” and “Middle (or Far) East” which imply that everything is viewed from the standpoint of Western Europe. There is also an increasing tendency for the terms “North” and “South” to be used to imply a particular stage of development. Usage of all such terms should be restricted to a geographical context only. Similarly, refer to “developing countries” rather than to “underdeveloped countries” or “the Third World”.

Further reading

Guidelines for reporting and writing about people with disabilities, 3rd ed. Lawrence, KS, Research and Training Center on Independent Living, 1990.

Miller C, Swift K. *The handbook of non-sexist writing*. London, The Women's Press, 1981.

Publication manual of the American Psychological Association, 4th ed. Washington, DC, American Psychological Association, 1994.

Spender D. *Man made language*, 2nd ed. London, Routledge & Kegan Paul, 1985:153–154.

16. Politically and legally sensitive topics

Occasionally, manuscripts contain statements that may cause offence to certain Member States, or references to geographical entities whose name or ownership is disputed. It would be unrealistic to try to provide a comprehensive list of sensitive topics, and editors need to be alert to potentially controversial issues. These may include, for example, terminology that implies a Euro-centric view of the world (Middle East, Far East), references to countries as underdeveloped, and suggestions that a particular government is corrupt. As a general rule, any text in a technical publication that describes the workings of, or criticizes, a particular government or national health service should have been cleared by the government concerned; this is the responsibility of the technical unit. Brief statements of this nature presented as examples from particular countries or as attributed views from other publications are generally acceptable, but editors should always be aware of the possibility of causing embarrassment to governments and should clear any doubtful text through LEG.

Political and geographical designations

In general, WHO follows United Nations practice in the field of geographical terminology. However, when situations are changing fast it may be impossible for editors to keep up with the latest developments. If editors have any doubts on the acceptability of a particular name or designation or the status of a country with respect to membership of WHO, they should check with LEG. (For quick reference, the Official List of Addresses contains a regularly updated list of Member States.)

When several Member States are referred to in the same context in tables, figures or text, and there is no obvious reason for grouping them together in a particular way (e.g. by perinatal mortality rate or in order of gross national product), they should be listed in alphabetical order.

Country, state, territory

All WHO publications carry a standard disclaimer regarding the designation of countries, territories, cities, areas, and their authorities, and the delimitation of their frontiers (see section 8). The term “country” is often construed as meaning a sovereign state, and territories not responsible for their international relations, e.g. Gibraltar,

should not be included in listings bearing the title “country”. Normally, the heading “country or area” can be used in table headings, etc. to cover such cases.

If anything is to be said about the status of non-self-governing territories, they should be referred to as “territories that are not responsible for the conduct of their international relations”; the words “colony” and “colonial” should not be used.

Maps

All maps used in WHO publications should follow United Nations practice with respect to place names and drawing of boundaries. A copy of the standard United Nations map of the world is with GRA. Maps for publication should be cleared through LEG if they have not been prepared by GRA or if editors are in any doubt about particular borders or areas.

Specific areas

The following are some examples of designations where care is needed.

Taiwan

WHO considers Taiwan as a province of China, under the jurisdiction of the Chinese Government in Beijing. If it is felt necessary to include data from Taiwan, it should be referred to as China (Province of Taiwan) or as Taiwan, China. WHO does not recognize the term Taiwan, Republic of China.

The Gulf

The term “Persian Gulf” is the standard geographical designation for the body of water lying between the Arabian peninsula and the Islamic Republic of Iran. It is recognized, however, that a number of states prefer the term Arabian Gulf, and in text referring specifically to these states the term “Gulf” may be used.

Occupied Territories

Any references to the Occupied Territories or Palestine should be referred to LEG for comment.

United Kingdom

The term United Kingdom refers to the Member State, but care should be taken in applying it to the legal system or health services, for which separate provisions exist in England and Wales, Northern Ireland, and Scotland. When the context is neither political nor legal, as in addresses and geographical descriptions, the relevant constituent part of the United Kingdom should be named (e.g. England, Scotland, Great Britain).

Trade marks and manufacturers' names

All WHO publications carry a disclaimer concerning the mention of proprietary products (see section 8). Nevertheless, to avoid the implication that WHO endorses or recommends a particular manufacturer's product (drug, pesticide, item of medical equipment) in preference to others, generic names should be used instead of trade names whenever possible. If a specific proprietary product must be named in a technical publication, it should be distinguished by an initial capital letter; it may be necessary to explain its inclusion if the context is particularly sensitive.

17. *Technical reports*

Manuscripts submitted for publication in the Technical Report Series should in general be edited in the usual way, in line with the guidelines set out elsewhere in the *WHO editorial style manual*. However, it should be borne in mind that most are the reports of WHO Expert Committees, Study Groups or Scientific Groups, whose members have in theory approved the final text at the time of the meeting. Major restructuring and substantial deletions or additions are therefore not possible. The secretary of the meeting will often be able to identify sections in the text or particular topics that provoked heated debate or lengthy discussion during the meeting and that as such may require cautious editing.

In general, expert groups review given fields of public health importance and make technical recommendations on subjects of interest to WHO. Their findings and recommendations are intended to assist national administrations and international health organizations in formulating their health policies and in establishing standards. This function, and any specific objectives identified in the introduction to the group's report, should be kept in mind during the editing of the text.

The following specific editorial points deserve particular attention; useful additional information can be found in the WHO Manual, section IX.1, Annex A.

Title of the report and expert group

The title of the report usually takes one of three forms:

1. EVALUATION OF CERTAIN FOOD ADDITIVES AND
CONTAMINANTS
Thirty-seventh report of the Joint FAO/WHO Expert Committee
on Food Additives
2. ENVIRONMENTAL HEALTH IN URBAN DEVELOPMENT
Report of a WHO Expert Committee
3. WHO EXPERT COMMITTEE ON DRUG DEPENDENCE
Twenty-seventh report

The proposed descriptive title of a report may be edited if necessary, but the name of the expert group itself cannot be changed after the meeting.

List of participants

The list of participants should be headed by the name of the expert group, irrespective of the form of the title of the report.

The names of members should be listed in alphabetical order, each followed by the title of the post held, the name of the institute/university department, and its geographical location. Titles, names of institutes, etc. should normally be given in English, but if no exact equivalent exists or in case of doubt, the name should be left in the original language. The Chairman, Vice-Chairman and Rapporteur should be identified.

The names of representatives of United Nations bodies and of nongovernmental and intergovernmental organizations in official relations with WHO should be listed under the heading "Representatives of other organizations".

Under the heading "Secretariat" only the secretary and any consultants and temporary advisers who assisted in the preparation of the meeting should normally be listed. Other names may be included only under the special circumstances detailed in the WHO Manual, section IX.1, Annex A, paragraph 160. Acknowledgements of any special contributions made by people not included in the list of participants may be included at the end of the report.

The secretary of the meeting is responsible for the correctness of the details given in the list of participants. However, additional information concerning members of expert advisory panels (i.e. all Expert Committee members and many members of Study Groups and Scientific Groups) can be obtained from the Office of Research Policy and Strategy Coordination (RPS).

Changes in titles, addresses, country names and WHO unit/division names that postdate the meeting should not be recorded.

Main text of the report

Controversial statements

Every Technical Report carries a disclaimer, stating that the report "contains the collective views of an international group of experts and does not necessarily represent the decisions or the stated policy of the World Health Organization". Nevertheless, care should be taken to rephrase statements that might prejudice the interests of the Organi-

zation or of Member States, for example overt criticism of named national governments or criticism relating to WHO's internal organization and to the functioning of specified technical units. Over-generalizations concerning particular professional or population groups and legally contentious issues should be avoided. In case of doubt about specific statements, LEG should be consulted.

Specifications

Specifications and similar requirements drafted by a meeting of experts are not necessarily endorsed by WHO and should not be described as WHO specifications unless they have been adopted by the World Health Assembly or the Executive Board. They may, however, be described as "specifications published by WHO".

Definitions

Definitions recommended by an expert group should not conflict with definitions previously published by WHO, unless the group considered that previous definitions should be superseded, when this should be explicitly stated. A phrase such as "the Expert Committee adopted the following definitions for the purposes of its report" may be useful when the status and scope of applicability of definitions are not clear.

Headings

Headings should be sufficiently numerous to separate clearly the various topics discussed and to facilitate reading, but excessive use of third-order and fourth-order headings should be avoided. A numbering system is normally used (1, 1.1, 1.2, 1.2.1, etc.) but should not be extended beyond third-order headings.

Recommendations

The recommendations of an expert group are often split into those directed to WHO and those directed to Member States. The former frequently need to be reworded so as to advise the Director-General on future programme activities without calling upon him to use WHO staff, services or funds in a specified way. Recommendations of the form "WHO should fund..." and "WHO should establish a programme for..." are not acceptable. Direct recommendations to other United Nations bodies and to nongovernmental organizations should be avoided by appropriate rewording, for example "WHO should encourage/collaborate with/urge UNICEF..." rather than "UNICEF should...".

References

References should be numbered consecutively throughout the text and should be listed together at the end of the report, before any annexes (which may have their own reference lists). Only if a single reference is cited in the report should it be given in a footnote. The bibliographic style to be followed is given in section 9.

Normally references to WHO unpublished documents should not be included; any relevant information they contain should be summarized in the report. However, if an unrestricted document, available on request to readers, contains important supplementary information that cannot be incorporated in the report, a reference to it may be included in the reference list, together with an indication as to how the document can be obtained.

Bibliographies

A bibliography is intended to provide the reader with a list of publications that may be consulted as background reading or for supplementary information. It should be reasonably representative of the relevant literature that exists in the languages of publication of the Technical Report Series (always English, French and Spanish, and sometimes Arabic, Chinese or Russian). If the list provided does not seem to fulfil these criteria, it may need to be modified, removed or retitled.

Annexes

Expert groups often annex supplementary information, details of technical procedures, and summary lists and tables to their reports; such material should have been approved by the group at the time of the meeting. All annexes should be referred to in the body of the report, with an indication of the reasons for their inclusion. Working papers and signed contributions may not be included as annexes.

18. **Useful reference books**

Dictionaries

Thompson D, ed. *The concise Oxford dictionary*, 9th ed. Oxford, Clarendon Press, 1995.

Critchley M, ed. *Butterworths medical dictionary*, 2nd ed. London, Butterworths, 1978.

Dorland's illustrated medical dictionary, 28th ed. Philadelphia, Saunders, 1994.

Collins English dictionary, 4th ed. London, HarperCollins, 1994.

Landau SI, ed. *International dictionary of medicine and biology*. New York, Wiley, 1986.

Stedman's medical dictionary, 26th ed. Baltimore, Williams & Wilkins, 1995.

Webster's new geographical dictionary. Springfield, Merriam-Webster, 1984.

Style manuals

Butler P. *The Economist style guide*. London, The Economist Books, 1991.

CBE Style Manual Committee. *Scientific style and format: the CBE style manual for authors, editors, and publishers*, 6th ed. Cambridge, Cambridge University Press, 1994.

Huth EJ. *Medical style and format: an international manual for authors, editors, and publishers*. Philadelphia, ISI Press, 1987.

Iverson C et al. *American Medical Association manual of style: a guide for authors and editors*, 9th ed. Baltimore, Williams & Wilkins, 1997.

Publication manual of the American Psychological Association, 4th ed. Washington, DC, American Psychological Association, 1994.

The Chicago manual of style: the essential guide for writers, editors, and publishers, 14th ed. Chicago, University of Chicago Press, 1993.

English usage, style and editing

A guide to writing for the United Nations. New York, United Nations, 1984.

Bishop C. *How to edit a scientific journal*. Philadelphia, ISI Press, 1984.

Burchfield RW, ed. *The new Fowler's modern English usage*. Oxford, Clarendon Press, 1996.

Butcher J. *Copy-editing: the Cambridge handbook for authors, editors and indexers*, 3rd ed. Cambridge, Cambridge University Press, 1992.

Carey GV. *Mind the stop: brief guide to punctuation*. Harmondsworth, Penguin Books, 1971.

Day RA. *How to write and publish a scientific paper*, 4th ed. Cambridge, Cambridge University Press, 1995.

Hart's rules for compositors and readers at the University Press Oxford, 39th ed. Oxford, Oxford University Press, 1983.

Huth EJ. *How to write and publish papers in the medical sciences*, 2nd ed. Baltimore, Williams & Wilkins, 1990.

Gowers E (rev. Greenbaum S, Whitcut J). *The complete plain words*, 3rd ed. London, HMSO, 1986.

Miller C, Swift K. *The handbook of non-sexist writing*. London, The Women's Press, 1981.

O'Connor M. *Editing scientific books and journals: an ELSE-Ciba Foundation guide for editors*. London, Pitman, 1978.

O'Connor M. *How to copyedit scientific books and journals*. Philadelphia, ISI Press, 1986.

Spender D. *Man made language*, 2nd ed. London, Routledge & Kegan Paul, 1985.

Strunk W, White EB. *The elements of style*, 3rd ed. New York, Macmillan, 1978.

Scientific, technical and medical terminology and usage

Armstrong Lowe D. *A guide to international recommendations on names and symbols for quantities and on units of measurement*. Geneva, World Health Organization, 1975 (Progress in standardization, No. 2).

Baron DN, ed. *Units, symbols, and abbreviations: a guide for biological and medical editors and authors*, 4th ed. London, Royal Society of Medicine Services, 1988.

Budavari S et al., eds. *The Merck index*, 12th ed. Whitehouse Station, NJ, Merck, 1996.

Cahn RS, Dermer OC. *Introduction to chemical nomenclature*, 5th ed. London, Butterworths, 1979.

Enzyme nomenclature, 1992: recommendations of the Nomenclature Committee of the International Union of Biochemistry and Molecular Biology on the nomenclature and classification of enzymes. San Diego, Academic Press, 1992.

Guidelines for reporting and writing about people with disabilities, 3rd ed. Lawrence, KS, Research and Training Center on Independent Living, 1990.

Heister R. *Dictionary of abbreviations in medical sciences*. Berlin. Springer Verlag, 1989.

International code of zoological nomenclature adopted by the 20th General Assembly of the International Union of Biological Sciences, 3rd ed. London, International Trust for Zoological Nomenclature, 1985.

International Nomenclature of Diseases. Geneva, CIOMS or CIOMS/WHO, 1979–1992.

Volume II, infectious diseases

Part 1, bacterial diseases

Part 2, mycoses

Part 3, viral diseases

Part 4, parasitic diseases

Volume III, diseases of the lower respiratory tract

Volume IV, diseases of the digestive system

Volume V, cardiac and vascular diseases

Volume VI, metabolic, nutritional, and endocrine disorders

Volume VII, diseases of the kidney, the lower urinary tract, and the male genital system

Volume VIII, diseases of the female genital system

International nonproprietary names (INN) for pharmaceutical substances, cumulative list no. 9. Geneva, World Health Organization, 1996.

International Statistical Classification of Diseases and Related Health Problems, tenth revision. Volume 1: tabular list; Volume 2: instruction manual; Volume 3: alphabetical index. Geneva, World Health Organization, 1992–1994.

ISO standards handbook: 2. Units of measurement, 2nd ed. Geneva, International Organization for Standardization, 1982.

IUPAC Commission on the Nomenclature of Inorganic Chemistry. *Nomenclature of inorganic chemistry*, 3rd ed. Oxford, Blackwell Scientific, 1990.

IUPAC Commission on the Nomenclature of Organic Chemistry. *Nomenclature of organic chemistry: sections A, B, C, D, E, F and H*, revised ed. Oxford, Pergamon, 1979.

Jablonski S. *Dictionary of medical acronyms and abbreviations*, 2nd ed. Philadelphia, Hanley & Belfus, 1993.

Copping LG, ed. *Pesticide index: an index of chemical, common and trade names of pesticides and related crop-protection products*, 3rd ed. Farnham, British Crop Protection Council, 1995.

Lapage SP, Sneath PH, eds. *International code of nomenclature of bacteria*. Washington, DC, American Society for Microbiology, 1992.

Liébecq C. *Biochemical nomenclature and related documents, a compendium*, 2nd ed. London, Portland Press (for the International Union of Biochemistry and Molecular Biology), 1992.

Manuila A, ed. *Progress in medical terminology*. Basel, Karger, 1981.

Marler EEJ. *Pharmacological and chemical synonyms*, 10th ed. Amsterdam, Elsevier, 1994.

Quantities, units and symbols, 2nd ed. London, The Royal Society, 1975 (amendment 1981).

Riegelman RK, Hirsch RP. *Studying a study and testing a test: how to read the medical literature*, 2nd ed. Boston, Little, Brown & Co., 1989.

Skerman VBD, McGowan V, Sneath PHA, eds. *Approved lists of bacterial names*, amended ed. Washington, DC, American Society for Microbiology, 1989.

Steen EB. *Baillière's abbreviations in medicine*, 5th ed. London, Baillière Tindall, 1984.

Swinscow TDV. *Statistics at square one*, 9th ed (revised). London, BMJ Publishing Group, 1996.

The Oxford dictionary for scientific writers and editors. Oxford, Clarendon Press, 1991.

The SI for the health professions. Geneva, World Health Organization, 1977 (updated and reprinted 1982).

Tomlin C, ed. *The pesticide manual: a world compendium*, 10th ed. Farnham, British Crop Protection Council, 1994.

Miscellaneous

Currency units. New York, United Nations, 1993 (Terminology bulletin no. 346).

Hunter B, ed. *The statesman's year-book: statistical, political and economic account of the states of the world for the year 1997–1998*. New York, St Martin's Press, 1997.

Annex 1

Spelling of words that commonly cause difficulty

The general rule in WHO is to follow the spelling given in the latest edition of *The concise Oxford dictionary*; where it gives alternative spellings or forms, use the one given first (e.g. fulfil, fulfill) or the one to which you are referred from an entry giving a variant spelling (e.g. "tire, US var. of tyre": use "tyre").

There are, however, a number of exceptions, owing to established usage within WHO and to the recommendations of international nomenclature-setting bodies in a variety of disciplines. This list therefore gives the preferred WHO spellings of many of the commonly encountered exceptions; it also includes *The concise Oxford dictionary* spellings that should be followed for many words that often cause doubt or difficulty, and gives some guidance on usage and choice of words.

A

above-mentioned	African Region (of WHO)
abridgement	after-care
absorbent	after-effect
absorption	age group
accommodation	agenda (treat as singular; pl. agendas)
accumulate	ageing
acknowledgement	aid (thing)
acquired immunodeficiency	aide (person)
syndrome (not...immune	AIDS (abbr. for acquired immuno-
deficiency...)	deficiency syndrome; see also
adaptation (not adaption)	human immunodeficiency virus)
adapter (person)	airborne
adaptor (electrical/mechanical)	air-conditioning
addendum (pl. addenda)	airflow
ad hoc	alga (pl. algae)
advice (n.)	align, alignment
advise (vb)	all right (not alright)
adviser (not advisor)	alternate, alternately (deputy or in
<i>Aedes</i> (not <i>Aëdes</i>)	turn)
aerial	alternative, alternatively (options)
aeroplane	aluminium
aesthetic	amfetamine
aforementioned	amino acids
aforesaid	amoebiasis
	among (not amongst)
	analyse

anaesthesia, anaesthetic	bibliographic (<i>not</i> bibliographical)
ancylostomiasis (<i>not</i> ankylo...)	biennial (every 2 years)
animal feed	biennium (<i>pl.</i> bienniums)
ankylose, ankylosing spondylitis	bimonthly, biweekly, biyearly
annex (n. & vb)	(ambiguous; avoid and rephrase
annul, annulled, annulling,	– every 2 months, etc. or twice a
annulment	month)
<i>Anopheles</i>	bioassay
anopheline mosquitos; anophelines	birth rate
antecedent	birth weight (<i>but</i> low-birth-weight
antedate	infants)
antemortem	blender (<i>not</i> blender)
antenatal	bloodborne
anthelminthic	blood cells
antibacterial	blood donor
anticoagulant	blood group (n.; <i>but</i> blood-grouping)
anticomplement	blood-meal
antiepileptic	blood pressure
anti-gas-gangrene	bloodstream
antimicrobial(s)	blood type (n.; <i>but</i> blood-typing)
antineoplastic	body weight
antioxidant(s)	boiling-point
antiserum, antisera	bona fide
any one (of several)	book-keeper
anyone (anybody)	borderline (n. & adj.)
apartheid	born (child)
apnoea, apnoeic	borne (carried)
appendix (<i>pl.</i> appendices, all senses)	bottle-fed, bottle-feed (vb), bottle-
arbovirus(es)	feeding
arc, arced, arcing (<i>not</i> arcked, etc.)	break down (vb)
artefact (<i>not</i> artifact)	breakdown (n. & adj.)
ascariasis	breastfed, breastfeed (vb),
audiovisual	breastfeeding
autoimmune	breast milk (<i>but</i> breast-milk
autoxidation	substitute)
	budget, budgeting, budgeted
B	bureaux
backward (adj.)	by-law
backwards (adv.)	bypass (n. & vb)
bacterium (<i>pl.</i> bacteria)	by-product
baseline (n. & adj.)	
bedbug	C
bednet	°C (no space)
befit, befitted	caesarean
benefit–risk (ratio)	caesium
benefit, benefited, benefiting	calliper (<i>not</i> caliper)
biannual (twice a year)	cannot (1 word)
biased	canvas (cloth)

canvass (solicit)	concomitant
carcass (<i>not</i> carcase)	connection (<i>not</i> connexion)
cardiovascular	consensus (<i>not</i> concensus)
case-finding	consummate (adj. & vb)
case history	continual (intermittent but frequent)
case-load	continuous (without a break)
case-study	contraindicate, contraindication
caster (person or machine)	conveyor
castor (oil)	cooperate
catalogue	coopt
census	coordinate
centimetre (cm)	cornerstone
centre, centring	corrigendum (<i>pl.</i> corrigenda)
century (20th, etc.)	cosponsor
cerebrospinal	cost-benefit (ratio)
cerebrovascular	cost-effectiveness (ratio)
Chagas disease	co-trimoxazole
checklist	cotton wool (<i>not</i> cottonwool)
check up (vb)	councillor (member of council)
check-up (n.)	counsellor (adviser)
cheque (banking)	counterproductive
chequered (flag, career)	countrywide (adj.)
childbearing	coverslip
childbirth	co-worker
child-spacing	criterion (<i>pl.</i> criteria)
<i>Chlamydia</i> (<i>but</i> chlamydiae, chlamydial)	cross-matching
classroom	cross-react, cross-reaction
coauthor	cross-reference
co-eluting	cross-resistance
coenzyme	cross-section
coexist, coexistence	curriculum (<i>pl.</i> curricula)
cofactor	curriculum vitae (<i>pl.</i> curricula vitae)
cold chain	cut-off point
cold-chain equipment	
coloration	
colorimeter	
colour	
combat, combated, combating, combative	D
commit, commitment	dare say (<i>not</i> daresay)
common sense (n.)	data (<i>plural</i> word; <i>sing.</i> datum)
common-sense (adj.)	data bank
commonsensical	database
complement, complementary (make complete)	day care
complexion	daytime
compliment, complimentary (praise; free as a compliment)	deadline
	death rate
	decision-maker, decision-making
	de facto
	defecate

- defence
 defensible, defensive
 dependant (n.)
 dependent (adj.)
 depository (person)
 depository (library, place)
 deprecate (disapprove)
 depreciate (belittle)
 dermatotropic (*not* dermatopic)
 desiccate
 device (n.)
 devise (vb)
 diarrhoea
 dilatation (medical)
 dilation (verbal)
 diphtheria
 Diptera
 dipterous (insects)
 Director-General
 disassemble (take to pieces)
 disc
 discoloration
 discolour
 discreet (unobtrusive)
 discrete (distinct, separate)
 disinsection (*not* disinsectization)
 diskette
 dispatch (*not* despatch)
 dissemble (conceal)
 dissociate (*not* disassociate)
 divers (several, various)
 diverse (different, divergent)
 Down syndrome (*not* Down's)
 Dr (*not* Dr.)
 dracunculiasis (= guinea-worm disease; *avoid* dracontiasis)
 draft, draftsman (of texts)
 draught, draughtsman (of drawings)
 drier (comp. adj. & machine)
 drinking-water (*but* International Drinking Water Supply and Sanitation Decade)
 drop-out (n.)
 drop out (vb)
 dysentery, dysenteric
 dyspnoea, dyspnoeic
- E**
 earmark (n. & vb)
 eastern Europe (geographical)
 Eastern Europe (political)
 Eastern Mediterranean Region (of WHO)
 Economic and Social Council (of the United Nations) (*not* ECOSOC)
 e.g. (no comma)
 elastomer, elastomeric
 electron microscope
 electron microscopy
 élite
 embarrassment
 employee
 enclose (*not* inclose)
 end-point
 end-product
 enquire (seek information)
 enrol, enrolled, enrolment
en route
 ensure (make sure; *compare* insure)
 enterocolitis
 enzyme-linked immunosorbent assay (ELISA) (*not* ...immuno-absorbent; *not* Elisa)
 erratum (*pl.* errata)
 estrogen
 estrous (adj.)
 estrus (n.)
 et al.
 etiology (*not* aetiology)
 et seq.
 European Region (of WHO)
 evaporating-dish
 everyday (adj.)
 every one (of several)
 everyone (everybody)
 excreta (treat as plural)
 Executive Director of UNICEF (*no* hyphen)
 extra-articular
 extrabudgetary
 extrahepatic
 extravascular

F

fact-finding
 faecal, faeces
fait accompli
falciparum malaria (but Plasmodium falciparum)
 fallout (n.)
 fall out (vb)
 false-positive reaction
 far-reaching
 favour, favourable
 federal (authorities etc; in general contexts)
 Federal (with specific titles & names)
 feed (for animals)
 feed back (vb)
 feedback (n.)
 feldsher
 fetal, fetus
 fibre (not fiber)
 fibroadenoma
 fibroma
 field test (n.)
 field-test (vb)
 field trial
 fieldwork
 Fig. ("as shown in Fig. 3")
 figure ("the figure above")
 filter
 filter-paper
 fine-tooth comb (not fine tooth-comb)
 finger-prick (n. & adj.)
 first aid (*but* first-aid kit, etc.)
 First World War (*not* World War I)
 flammable (*prefer to* inflammable)
 flow-chart
 flowsheet
 focus, focused, focusing
 follow up (vb)
 follow-up (n. & adj.)
 Food and Agriculture Organization of the United Nations
 (not...Agricultural; *do not omit* "of the United Nations")
 foodborne
 food-chain
 foodstuff (for people)
 foothold
 footnote
 forbear (vb; refrain from)
 forebear (n; ancestor)
 forebrain
 forego (precede)
 foregone (conclusion)
 foreseeable
 for ever (for always)
 forever (continually)
 foreword (preface)
 forgo (go without)
 format, formatted, formatting
 (*compare* combat)
 forum (pl. forums)
 forward(s) (onward)
 fourfold (*but* 4-fold)
 framboesia
 freezing-point
 fresh water (n.)
 freshwater (adj.)
 front-line health workers
 fulfil, fulfilled, fulfilment
 further, furthest (*prefer to* farther, farthest)

G

γ -isomer (not gamma isomer)
 g (gravity; e.g. 3.5g)
 gas-gangrene
 gastroenteritis
 gastrointestinal
 gauge (n. & vb; not guage)
 General Service category (of staff)
 genetic (*not* genetical)
 genitourinary
 genus (pl. genera)
 geographical (*not* geographic)
 geriatric (concerns health & welfare of the elderly; *not* a synonym for aged or senile)
 giardiasis
 glasnost
 goitre (*not* goiter)
 gonococcal
 gonococcus (*not* Gonococcus; *see* *Neisseria*)

- good will (benevolent interest)
 goodwill (of firm)
 gorilla
 governing body (*but* Governing Body of ILO)
 government (in general contexts)
 Government (of particular country)
 gram (g)
 Gram-positive/negative
 granddaughter, grandchild, etc.
 grass roots (n.)
 grass-roots (adj.)
 gray (radiation unit; symbol = Gy)
 green belt
 grey (colour; *not* gray)
 groundnut
 groundwater (n. & adj.)
 growth chart
 growth curve
 growth rate
 guerrilla
 guidelines
 guinea-pig
 guinea-worm disease
 (= dracunculiasis)
 gynaecological, gynaecology
 gynaecomastia
- H**
- haematocrit (*use* erythrocyte volume fraction)
 haematological, haematology
 haematopoietic (*not* haemopoietic)
 haemoglobin
 haemorrhage, haemorrhagic
 haemorrhoids
 half-hour (*but* half an hour)
 half-yearly
 handbook
 handful(s)
 handicap, handicapped, handicapping
 harass, harassment
 headquarters (of WHO etc.)
 health for all (*but* Global Strategy for Health for All)
 health-for-all (adj.)
- helminth, helminthiasis
 herpesvirus(es)
 heterogeneity, heterogeneous (*not* heterogenous)
 highlight
 HIV (abbr. for human immunodeficiency virus)
 homoeopath, homoeopathy
 homoeostasis
 homogeneity, homogeneous(ness) (uniform; of the same kind)
 homogenetic, homogenous, homogeny (of common origin or descent)
 homogenize (blend; make homogeneous)
 honorarium (*pl.* honorariums)
 honorary
 honour, honourable
 horsepower
 hotbed (agriculture & intrigue)
 housefly
 however (between commas when it means nevertheless)
 HTLV-III (deprecated abbr. for virus associated with AIDS; *prefer* HIV)
 human immunodeficiency virus (recommended English vernacular name for virus associated with AIDS; abbr. = HIV)
 human T-cell lymphotropic virus type III (deprecated; *prefer* human immunodeficiency virus)
 humorous
 humour
 hydatid cyst, hydatidosis
 hydrocele
 hydrogen-ion concentration (*but* hydrogen ion when used as a noun)
 hyperactive, hyperactivity
 hyperthermia (excessively high body temperature)
 hypoactive, hypoactivity
 hypomyeloid
 hypothermia (excessively low body temperature)

I

i.e. (no comma)
 illegible
 ill effect
 ill-health
 immunochemistry
 immunodiagnosis
 immunological (not immunologic)
 immunomodulation
 immunosuppressive
 impracticable (impossible in practice)
 impractical (not practical)
 inasmuch as (2 words)
 indexes (of texts etc.)
 indices (mathematical, scientific)
 ineligible
 infarct, infarction (of myocardium)
 inflection (not inflexion)
 infraction (of rules)
 infrared
 infrastructure (does not normally need a plural)
 innocuous
 inoculate, inoculation
 inpatient
 input
 inquire, inquiry (of investigation, especially of official nature)
 insanitary (not unsanitary)
 in-service training
 insignia (plural word)
in situ
 in so far as (4 words)
 install, installed, instalment
 instil, instilled, instillation
 insure (against risk; compare ensure)
 interact, interaction
 interagency
inter alia
 inter-American
 Inter-American Development Bank
 intercostal
 intercountry
 interdepartmental
 intergovernmental
 International Labour Office (Secretariat)

International Labour Organisation (not ...Organization)
 interracial
 interregional
 interrelated, interrelationship
 intersectoral (not intersectorial)
 intervertebral
intifada
 into (1 word)
 intra-abdominal
 intracutaneous
 intradermal
 intramural
 intramuscular
 intraocular
 intraregional
 intrauterine
 intravascular
 intravenous
in vitro
in vivo
 iodinated
 iodized
 iron deficiency anaemia
 irradiated
 ischaemia, ischaemic
 isolatable (not isolable)

J

joule (J)
 Jr (= Junior)
 judgement (*but* Judgments of Administrative Tribunal of ILO)
 juxta-articular

K

Kellogg Foundation
 kerosene
 khat
 kilogram (kg)
 kilojoule (kJ)
 kilometre (km)
 kilowatt (kW)
 knee-jerks
 know-how
 kwashiorkor

L

label, labelled, labelling
labour, laborious
labour-intensive
lacrimal, lacrimate
laissez-faire
laissez-passer (invariable sing. & pl.)
large-scale (adj.; *but* on a large scale)
larva (pl. larvae)
last-mentioned
Latin American (n. & adj.)
lay out (vb)
layout (n.)
learned (erudite)
least developed countries
leptospirosis
leukaemia
leukocyte
leukopenia
liaison
licence (n.)
license (vb)
life cycle
life expectancy
lifelong
life span
lifestyle
life-support (adj.)
lifetime
lightening (the sky is)
lightning (the sky is full of)
liquefaction (*not* liquifaction)
liquefy (*not* liquify)
litre (symbol = l, *but* prefer litre where ambiguity may arise)
live birth
liveborn (adj.)
livestock
loath (adj.)
loathe (vb)
logbook
logistic (*not* logistical)
long-distance (adj.; *but* at a long distance)
long-term (adj.; *but* in the long term)
loophole
louse-borne

low back pain
low-cost (adj.; *but* at low cost)
lump-sum reimbursement (*but* as a lump sum)
lying-in hospital (*but* the mother was lying in)
lymphadenoma
lymphadenopathy
lymph glands
lymph nodes

M

macroeconomics
man-hours, man-months, etc.
(*prefer* work-hours, work-months, etc.)
man-made
manoeuvre, manoeuvrable (*not* maneuver, maneuverable)
marijuana (*not* marihuana)
matrix (pl. matrices)
may be (vb)
maybe (perhaps)
meantime
media (plural word; *sing.* medium)
media (pl. mediae) of blood vessel
Mediterranean
melting-point
Member countries
Member States (of WHO etc.)
members of the Executive Board
(persons designated by Member States)
memorandum (pl. memoranda)
meningococcal
meningococcus (*not* Meningococcus)
meningoencephalitis
meter (measuring device)
metre (unit of length; symbol = m)
microbiological, microbiology
microcomputer, microprocessor
microculture
microeconomics
microfiche (pl. microfiches)
micrometer (measuring device)
micrometre (unit of length)
microorganism
midday

millennium	newborn (adj. & abstract sing. n.; avoid newborns – prefer neo- nates)
millimetre (mm)	N-nitroso compounds (always italic cap.)
misspell, misspelt (not misspell, misspelled)	nobody
molarity (deprecated; use mol/litre)	non- (prefix; see footnote ¹ for use of the hyphen)
mollusc (not mollusk)	noncommittal
molluscicide (not molluscocide)	noncommunicable diseases
mortality rate	nonconformist
mosquitos (not mosquitoes)	nondescript
mould, mouldy (not mold, moldy)	nonentity
mouthwash	none the less (not nonetheless)
Mr	nongovernmental organizations
Mrs	nonplus
Ms	non-profit-making
MS, MSS (manuscript, sing. & pl.)	nonproprietary
mucopurulent	non-self-governing
mucous (adj.)	nonsense
mucus (n.)	nonserial
Muhammad (not Mahomet)	non-wage-earning
multidrug	no one (no hyphen)
multilateral	normality (not normalcy)
multinational	North (as in North–South dia- logue)
multipurpose	north-east, north-eastern
Muslim (not Moslem)	north-west, north-western
	nought (zero)
N	O
naive (not naïve)	oblivious of (not to)
nationwide	occur, occurred, occurrence
naught (nothing; see also nought)	oculomotor
near by (adv.)	odour, odourless (but odorous, odoriferous)
nearby (adj.)	oedema, oedematous
needle-stick (n. & adj.)	oesophageal, oesophagus
<i>Neisseria gonorrhoeae</i> (the gonococ- cus)	offence (not offense)
<i>Neisseria meningitidis</i> (the meningo- coccus)	offensive
neonatal	offprint
neurobiology	offset
neuroendocrinological, neuroendo- crinology	offshore
neuron	
neurophysiology	
nevertheless (1 word)	

¹ In general, follow *The concise Oxford dictionary*, whose tendency is to include a hyphen after the prefix. Exceptions to this rule are listed above, including certain words for which the unhyphenated form has been established as the preferred usage within WHO.

old age; old-age pension
 omission
 onchocerciasis
 oncogenic
 oncoming (traffic, etc.)
 one-third
 ophthalmic, ophthalmology (*not*
 ophthalmic or ophthalmic, etc.)
 oral rehydration salts, oral rehydra-
 tion solution
 ordinance (legal instrument)
 ordnance (gunnery and, in UK,
 mapping)
 Organization (always cap. when
 referring to WHO, but
 Organisation in International
 Labour Organisation and
 Organisation for Economic Co-
 operation and Development)
 organophosphorus compounds
 (*not* ...phorous)
 organotin
 orient, oriented (*not* orientate,
 orientated)
 orthopaedic(s)
 osteoarthritis, osteoarthrosis
 out-of-date (adj.; *but* it is out of
 date)
 outpatient
 output
 overall (adj., adv. & n.)
 overemphasize
 overestimate
 overpopulated
 overproduction
 override
 overrule
 overseas (*not* oversea)
 oversee
 oversight (supervision & inadvert-
 ence)
 oversimplify, oversimplification
 overuse
 overview (*prefer* survey, outline,
 review)

P

P values (cap. ital.)
 paediatric(s)
 Pan American Health Organization
 Pan American Sanitary Bureau
 panic, panicked, panicky
 parallel, paralleled, paralleling
 paralyse
 parenteral (introduced other than
 by the alimentary tract)
 participate in (*not* at or to)
 particulate matter
 pasteurize (no cap.)
 pathognomic
 payroll
 pendant (n.)
 pendent (adj.)
 per annum
 per capita
 per cent (*but* % is usual)
 percentage
 per diem, per diem allowance (no
 hyphen)
 perestroika
 perinatal
 periodontal
 periurban
per se
 Petri dish
 phagocytose (*not* phagocytize)
 pharmacogenetics
 pharmacognosy
 pharmacopoeia
 phenomenon (*pl.* phenomena)
 phosphorous (adj.)
 phosphorus (n.)
 physicochemical
 pipeline
 pneumococcus, pneumococcal (*not*
Pneumococcus)
 policy-maker, policy-making
 polyclinic
 postbasic
 postmortem (*but* examine the body
 post mortem)
 postnatal
 postpartum
 post-session

post-test
 post-treatment
 postwar (adj. only)
 practice (n.)
 practise (vb)
 précis, précis-writer
 preconceive, preconception
 precondition
 predominant(ly) (*not*
 predominate(ly))
 pre-eclampsia
 pre-eminent
 pre-empt, pre-emption, pre-emptive
 preinvestment
 prenatal
 prerequisite
 preschool child(ren)
 presently (in a short while; now =
 at present)
 pre-session
 pretest
 pretreatment
 prewar (adj. only)
 prima facie
 primary health worker (*prefer*
 community health worker)
 primary-school child(ren)
 principal (chief; n. & adj.)
 principle (rule of conduct)
 printout (*but* print out the results)
 Professional (category of staff)
 progestogen (*not* progestagen)
 program (computers *only*)
 programme (all non-computer uses)
 proofreader
 propellant (n.)
 propellent (adj.)
 provenance (*not* provenience)
 psychoactive
 psychosocial
 psychotropic

Q

Q fever
 quantify (*not* quantitate)
 questionnaire
 queue, queued, queuing

R

raccoon (*not* raccoon)
 radioactive, radioactivity
 radiodiagnosis
 radiofrequency
 radioimmunoassay
 radioisotope
 radionuclide
 radiopacity, radiopaque
 radiotherapy
 rainwater (n. & adj.)
raison d'être
 rarefy, rarefaction (*not* rarify,
 rarification)
 readmit
 reagent
 reallocate
 reappear
 reassess
 reclassification, reclassify
 recover (regain possession; *but*
 re-cover = cover anew)
 recreation (leisure; *but* re-create =
 create anew)
 redefine
 re-edit
 re-educate
 re-employ
 re-enter, re-entry
 re-establish
 re-examine
 reflection (*not* reflexion)
 reform (amend for the better; *but*
 re-form = form again)
 regime (*not* régime)
 regimen (therapeutic)
 region (no cap. in generic or non-
 formal references: "in this
 region", "in three WHO regions")
 Region (of WHO) (cap. in specific
 or formal references: "in the
 European Region")
 Region of the Americas (of WHO;
 not American Region)
 Regional Committee for ...
 (specific region)
 regional committee (general)

Regional Director (same practice as for "Region/region" regarding caps)	<i>Salmonella</i> , salmonellae
Regional Office (same practice as for "Region/region" regarding caps)	salt water (n.)
regroup	saltwater (adj.)
reimplantation	salutary
reinfect, reinfection	sanatorium (not sanitarium; pl. sanatoria)
reinoculate, reinoculation	savanna (but Savannah, Georgia)
reinstate	sceptic, sceptical (not skeptic, skeptical)
reinsure	<i>Schistosoma</i> , schistosome, schistosomiasis
rendezvous (no hyphen)	schoolchild(ren) (but primary-school children, secondary-school children)
reopen	schoolteacher (but primary-school teacher, secondary-school teacher)
reorganize	seabed
repairable (can be mended)	sea level
reparable (loss can be made good)	seat-belt
repellent (n. & adj.)	seawater (n. & adj.)
re-route	Second World War (not World War II)
resign (give up office; but re-sign = sign again)	Secretariat (cap. only when referring to the organs of WHO: Health Assembly, Executive Board, Secretariat)
resister (person)	secretariat (in all generic references to staff)
resistor (electrical)	Secretary-General (of United Nations)
respect (with respect to, in respect of)	sectoral
restructure	self-care
résumé	semi-arid
rethink	semiquantitative
retreat (withdraw; but re-treat = treat again)	septic (tank, etc.)
reuse (n. & vb)	serodiagnosis
rewrite	seronegative, seronegativity
ricketts	seropositive, seropositivity
<i>Rickettsia</i> , rickettsiosis (sing.), rickettsioses, rickettsial	set back (vb)
rigor (pathology)	set-back (n.)
rigorous	set up (vb)
rigour (strictness)	set-up (n.)
risk-benefit (ratio)	sew, sewed, sewn (with thread)
role (not rôle)	sewage (the substance)
roll-call, roll-call vote	sewerage (the works)
rotavirus(es)	share-cropper
	shelf-life
	shellfish
S	
sac (biol.)	
saccharin (the sweetener)	
saccharine (sweet, sweetly polite)	
sack (bag, etc.)	

shiftwork	straight away (at once)
shift worker	straightforward
<i>Shigella</i> , <i>shigellae</i> (pl.)	strait-jacket
shigellosis	strait-laced
short-term (adj.; <i>but</i> in the short term)	strait(s) (narrow passage of water)
sickbed	stratum (pl. strata)
sick-leave	<i>Streptococcus</i> , streptococci (pl.), streptococcal
sickle-cell anaemia	stumbling-block
side-effects	subclause
siphon (<i>not</i> syphon)	subcommittee
skilful	subcutaneous
sleeping sickness	subdivide, subdivision
smallpox	subitem
small-scale (adj.; <i>but</i> on a small scale)	sublethal
snake-bite	subparagraph
soakaway	subprogramme
so-called (adj.; <i>but</i> it is properly so called)	subregion, subregional
sociocultural	sub-Saharan
socioeconomic	sub-Saharan
some time (at some time)	subsection
sometime (formerly)	subsoil
sometimes	subsystem
sorption	subtitle
South-East Asia Region (of WHO)	subtotal
south-east, south-eastern	subtropical, subtropics
Southern blot	subunit
south-west, south-western	sugar-beet
sow, sowed, sown (seeds etc.)	sugar-cane
spectrum (pl. spectra)	sulfa...
spelt (past participle)	sulfi...
spoonful(s)	sulfo...
starting-point	sulfur (<i>not</i> sulphur)
State (n.; country (when specified), Member States, and when the word is part of official title of a federated State: "New York State"; <i>but</i> "13 states of the United States of America")	supersede (<i>not</i> supercede)
state (used adjectivally and when referring generally to federated units: "the Indian states")	supervisor (<i>not</i> superviser)
stationary (at rest)	surveillance
stationery (paper etc.)	syllabus (pl. syllabuses)
stillbirth (n.)	sylvan (<i>not</i> silvan)
stillborn (adj. only)	synthesize (<i>not</i> synthesize)
stockpile (vb)	syringe
storey (of building)	
	T
	Table ("the data in Table 1")
	table ("the table below")
	target, targeted, targeting
	teamwork
	teaspoon, teaspoonful(s)
	template (<i>not</i> templet)

teratogen, teratogenic
 test-tube
 textbook
 text processing, text processor
 thermonuclear
 thermostable, thermostability
 through (*not thru*)
 tick-borne
 time-limit
 time-scale
 timetable
 title page
 ton (should be preceded by "long"
 or "short")
 tonne (*not* "metric ton" or "metric
 tonne")
 topsoil
 total (vb), totalled, totalling
 towards (*not toward*)
 traffic, trafficker, trafficking
 tranquilize, tranquilizer
 transnational
 transplacental
 travel, travelled, travelling
Trypanosoma, trypanosome,
 trypanosomiasis, trypanosomal
 tsetse fly
 tumefaction, tumefy
 tumour, tumorigenic
 turn out (vb)
 turnout (n.)
 turn over (vb)
 turnover (n.)
 twofold (*but* 2-fold)
 two-thirds
 typesetting
 tyre (n.)

U

ultrafiltration
 ultra-low-volume formulations
 (abbr. = ULV)
 ultramicroscopic
 ultraspecific
 ultraviolet
 unbiased

uncooperative
 uncoordinated
 underdeveloped
 underestimated
 undergraduate
 underprivileged
 underrate
 underserved (*insufficiently served*)
 underuse
 underutilize
 under way (*not underway*)
 underweight (adj.)
 undeserved (*unmerited*)
 unequivocal, unequivocally (*not*
 unequivocable, unequivocably)
 UNESCO (caps; *but* Unesco Press)
 UNICEF (caps)
 United Kingdom (*not UK*)
 United Nations (*not UN*)
 update
 up-to-date (adj.; *but it is up to date*)
 urine analysis (*not urinalysis*)

V

Vacutainer (trade name, requires
 cap.)
 vaporize, vapour, vaporizer
 Vaseline (trade name, requires cap.;
prefer petroleum jelly)
 vasodilator
 vasomotor
 vector-borne
 venepuncture (*not venipuncture*)
 venesection
 ventrolateral
 via
 vice-chairman, etc.
 vice versa
 vigorous, vigour
 viral
 virile
vis-à-vis
 vitamin A, etc. (*not* Vitamin A,
 etc.)
 viva voce
 viz. (*prefer* namely)
 vulvovaginitis

W

wage-earner, wage-earning
wagon (not waggon)
waive (the rules; *see also* wave)
waiver (n.; relinquishing a right)
wastewater
water-bath
waterborne
watercourse
waterworks
wave (the flag; *see also* waive)
wavelength
waver (vb; falter)
weal (not wheal)
weekday
weekend
well-baby clinic
well-being
well-water
western (geographical)
Western (political and cultural)
Western Pacific Region (of WHO)
wet-nurse
while (not whilst)
whooping cough (*prefer* pertussis)
-wide (as adjectival and adverbial
suffix, no hyphen: “worldwide”)
widespread (adj.)
wildlife
wilful
-wise (as suffix, avoid if possible)
word processing, word processor
work-hours
workload
work-months
workplace
workshop
World Health Assembly (may be
shortened to Health Assembly,
not to Assembly)
World Health Organization (not
Organisation)
worldwide (adj. & adv.)
worthwhile (adj.; *but* it is well
worth while)

X Y Z

X-ray (not x-ray)
xerography
xerophilic
xerophthalmia (not xerophtalmia or
xerophthalmia)
Xerox (trade name, requires cap.)
yearbook
yellow fever
zoonosis, zoonoses (pl.) (not zöö...)
zoonotic

Annex 2

Member States of the United Nations, members of specialized agencies or parties to the Statute of the International Court of Justice¹

<i>Short name</i>	<i>Full name</i>	<i>Adjective/People</i>	<i>Capital city</i>
Afghanistan	the Islamic State of Afghanistan	Afghan	Kabul
Albania	the Republic of Albania	Albanian	Tirana
Algeria	the People's Democratic Republic of Algeria	Algerian	Algiers
Andorra	the Principality of Andorra	Andorran	Andorra-La-Vella
Angola	the Republic of Angola	Angolan	Luanda
Antigua and Barbuda	Antigua and Barbuda	of Antigua and Barbuda	St John's
Argentina	the Argentine Republic	Argentine	Buenos Aires
Armenia	the Republic of Armenia	Armenian	Yerevan
Australia	Australia	Australian	Canberra
Austria	the Republic of Austria	Austrian	Vienna
Azerbaijan	the Azerbaijani Republic	Azerbaijani	Baku
Bahamas (the)	the Commonwealth of the Bahamas	Bahamian	Nassau
Bahrain	the State of Bahrain	Bahraini	Manama
Bangladesh	the People's Republic of Bangladesh	(of) Bangladesh, a Bangladeshi	Dhaka
Barbados	Barbados	Barbadian	Bridgetown
Belarus	the Republic of Belarus	Belarusian	Minsk
Belgium	the Kingdom of Belgium	Belgian	Brussels
Belize	Belize	Belizean	Belmopan

¹ This annex is valid as at 22 January 1998, but CTT or LEG should be consulted for information on more recent changes and additions. The listing of capital cities is provided for general information purposes; however, it cannot be relied upon as authoritative. The United Nations, which is the source of information for this annex, does not maintain an official list of capital cities.

Short name	Full name	Adjective/People	Capital city
Benin	the Republic of Benin	Beninese	Porto-Novo
Bhutan	the Kingdom of Bhutan	Bhutanese	Thimphu
Bolivia	the Republic of Bolivia	Bolivian	Sucre ¹
Bosnia and Herzegovina	Bosnia and Herzegovina	(of) Bosnia and Herzegovina	Sarajevo
Botswana	the Republic of Botswana	(of) Botswana	Gaborone
Brazil	the Federative Republic of Brazil	Brazilian	Brasília
Brunei Darussalam	Brunei Darussalam	(of) Brunei Darussalam	Bandar Seri Begawan
Bulgaria	the Republic of Bulgaria	Bulgarian	Sofia
Burkina Faso	Burkina Faso	(of) Burkina Faso	Ouagadougou
<i>Burma: see Myanmar</i>			
Burundi	the Republic of Burundi	(of) Burundi, Burundian	Bujumbura
Cambodia	the Kingdom of Cambodia	Cambodian	Phnom Penh
Cameroon	the Republic of Cameroon	Cameroonian	Yaoundé
Canada	Canada	Canadian	Ottawa
Cape Verde	the Republic of Cape Verde	Cape Verdean	Praia
Central African Republic (the)	the Central African Republic	of the Central African Republic, Central African	Bangui
Chad	the Republic of Chad	Chadian	N'Djamena
Chile	the Republic of Chile	Chilean	Santiago
China	the People's Republic of China	Chinese	Beijing
Colombia	the Republic of Colombia	Colombian	Bogotá
Comoros (the)	the Islamic Federal Republic of the Comoros	Comorian	Moroni
Congo (the)	the Republic of the Congo	Congolese	Brazzaville
Cook Islands	the Cook Islands	of the Cook Islands	Avarua
Costa Rica	the Republic of Costa Rica	Costa Rican	San José
Côte d'Ivoire	the Republic of Côte d'Ivoire	Ivorian	Yamoussoukro ²

¹ Seat of Government: La Paz.² Seat of Government: Abidjan.

Short name	Full name	Adjective/People	Capital city
Croatia	the Republic of Croatia	Croatian	Zagreb
Cuba	the Republic of Cuba	Cuban	Havana
Cyprus	the Republic of Cyprus	Cypriot	Nicosia
Czech Republic (the)	the Czech Republic	Czech	Prague
Democratic People's Republic of Korea (the)	the Democratic People's Republic of Korea	of the Democratic People's Republic of Korea	Pyongyang
Democratic Republic of the Congo (the)	the Democratic Republic of the Congo	of the Democratic Republic of the Congo, Congolese	Kinshasa
Denmark	the Kingdom of Denmark	Danish, a Dane	Copenhagen
Djibouti	the Republic of Djibouti	(of) Djibouti, a Djiboutian	Djibouti
Dominica	the Commonwealth of Dominica	(of) Dominica	Roseau
Dominican Republic (the)	the Dominican Republic	Dominican	Santo Domingo
Ecuador	the Republic of Ecuador	Ecuadorian	Quito
Egypt	the Arab Republic of Egypt	Egyptian	Cairo
El Salvador	the Republic of El Salvador	Salvadoran	San Salvador
Equatorial Guinea	the Republic of Equatorial Guinea	(of) Equatorial Guinea	Malabo
Eritrea	Eritrea	Eritrean	Asmara
Estonia	the Republic of Estonia	Estonian	Tallinn
Ethiopia	Federal Democratic Republic of Ethiopia	Ethiopian	Addis Ababa
Fiji	the Republic of Fiji	of Fiji	Suva
Finland	the Republic of Finland	Finnish, a Finn	Helsinki
France	the French Republic	French, a French- man, the French	Paris
Gabon	the Gabonese Republic	Gabonese	Libreville
Gambia (the)	the Republic of the Gambia	Gambian	Banjul
Georgia	Georgia	Georgian	Tbilisi

Short name	Full name	Adjective/People	Capital city
Germany	the Federal Republic of Germany	German	Berlin ¹
Ghana	the Republic of Ghana	Ghanaian	Accra
Greece	the Hellenic Republic	Greek	Athens
Grenada	Grenada	Grenadian	St George's
Guatemala	the Republic of Guatemala	Guatemalan	Guatemala City
Guinea	the Republic of Guinea	Guinean	Conakry
Guinea-Bissau	the Republic of Guinea-Bissau	(of) Guinea-Bissau	Bissau
Guyana	the Republic of Guyana	Guyanese	Georgetown
Haiti	the Republic of Haiti	Haitian	Port-au-Prince
Holy See (the) ²	the Holy See	of the Holy See	
Honduras	the Republic of Honduras	Honduran	Tegucigalpa
Hungary	the Republic of Hungary	Hungarian	Budapest
Iceland	the Republic of Iceland	Icelandic, an Icelander	Reykjavik
India	the Republic of India	Indian	New Delhi
Indonesia	the Republic of Indonesia	Indonesian	Jakarta
Iran (Islamic Republic of) ³	the Islamic Republic of Iran	Iranian	Teheran
Iraq	the Republic of Iraq	Iraqi	Baghdad
Ireland	Ireland	Irish, the Irish, an Irishman	Dublin
Israel	the State of Israel	Israeli	
Italy	the Italian Republic	Italian	Rome
<i>Ivory Coast (the): see Côte d'Ivoire</i>			
Jamaica	Jamaica	Jamaican	Kingston
Japan	Japan	Japanese	Tokyo
Jordan	the Hashemite Kingdom of Jordan	Jordanian	Amman

¹ Seat of Government: Bonn/Berlin.

² In United Nations documents the term "Holy See", not "Vatican City State", should be used, except in texts concerning the International Telecommunication Union and the Universal Postal Union – the two organizations in which the latter term is the accepted designation.

³ For purposes other than alphabetical lists or name-plates, the following form should be used: the Islamic Republic of Iran.

Short name	Full name	Adjective/People	Capital city
Kazakhstan	the Republic of Kazakhstan	Kazakh	Almaty
Kenya	the Republic of Kenya	Kenyan	Nairobi
Kiribati	Kiribati	of Kiribati	Tarawa
<i>Korea: see Democratic People's Republic of Korea (the), and Republic of Korea (the)</i>			
Kuwait	the State of Kuwait	Kuwaiti	Kuwait
Kyrgyzstan	the Kyrgyz Republic	Kyrgyz	Bishkek
Lao People's Democratic Republic (the)	the Lao People's Democratic Republic	Lao, a Lao, the Lao ¹	Vientiane
Latvia	the Republic of Latvia	Latvian	Riga
Lebanon	the Lebanese Republic	Lebanese	Beirut
Lesotho	the Kingdom of Lesotho	(of) Lesotho	Maseru
Liberia	the Republic of Liberia	Liberian	Monrovia
Libyan Arab Jamahiriya (the)	the Socialist People's Libyan Arab Jamahiriya	Libyan	Tripoli
Liechtenstein	the Principality of Liechtenstein	(of) Liechtenstein	Vaduz
Lithuania	the Republic of Lithuania	Lithuanian	Vilnius
Luxembourg	the Grand Duchy of Luxembourg	(of) Luxembourg	Luxembourg
<i>Macedonia: see The former Yugoslav Republic of Macedonia</i>			
Madagascar	the Republic of Madagascar	Malagasy	Antananarivo
Malawi	the Republic of Malawi	Malawian	Lilongwe
Malaysia	Malaysia	Malaysian	Kuala Lumpur
Maldives	the Republic of Maldives	Maldivian	Malé
Mali	the Republic of Mali	Malian	Bamako
Malta	the Republic of Malta	Maltese	Valletta
Marshall Islands (the)	the Republic of the Marshall Islands	of the Marshall Islands, Marshallese	Majuro
Mauritania	the Islamic Republic of Mauritania	Mauritanian	Nouakchott
Mauritius	the Republic of Mauritius	Mauritian	Port Louis
Mexico	the United Mexican States	Mexican	Mexico City

¹ "Lao" is invariable, both as an adjective and as a noun, in English, French and Spanish.

Short name	Full name	Adjective/People	Capital city
Micronesia (Federated States of) ¹	the Federated States of Micronesia	of the Federated States of Micronesia	Palikir
<i>Moldova: see Republic of Moldova (the)</i>			
Monaco	the Principality of Monaco	Monegasque	Monaco
Mongolia	Mongolia	Mongolian	Ulaanbaatar
Morocco	the Kingdom of Morocco	Moroccan	Rabat
Mozambique	the Republic of Mozambique	Mozambican	Maputo
Myanmar	the Union of Myanmar	of Myanmar	Yangon
Namibia	the Republic of Namibia	Namibian	Windhoek
Nauru	the Republic of Nauru	Nauruan	Yaren
Nepal	the Kingdom of Nepal	Nepalese	Kathmandu
Netherlands (the)	the Kingdom of the Netherlands	Netherlands, a Netherlander	Amsterdam ²
New Zealand	New Zealand	(of) New Zealand, a New Zealander	Wellington
Nicaragua	the Republic of Nicaragua	Nicaraguan	Managua
Niger (the)	the Republic of the Niger	(of the) Niger	Niamey
Nigeria	the Federal Republic of Nigeria	Nigerian	Abuja
Niue	the Republic of Niue	of Niue	Alofi
Norway	the Kingdom of Norway	Norwegian	Oslo
Oman	the Sultanate of Oman	Omani	Muscat
Pakistan	the Islamic Republic of Pakistan	(of) Pakistan, a Pakistani	Islamabad
Palau	the Republic of Palau	of Palau	Koror
Panama	the Republic of Panama	Panamanian	Panama City
Papua New Guinea	Papua New Guinea	(of) Papua New Guinea, a Papua New Guinean	Port Moresby
Paraguay	the Republic of Paraguay	Paraguayan	Asunción
Peru	the Republic of Peru	Peruvian	Lima

¹ For purposes other than alphabetical lists and name-plates, the following form should be used:
the Federated States of Micronesia.

² Seat of Government: The Hague.

Short name	Full name	Adjective/People	Capital city
Philippines (the)	the Republic of the Philippines	Philippine, a Filipino	Manila
Poland	the Republic of Poland	Polish, a Pole	Warsaw
Portugal	the Portuguese Republic	Portuguese	Lisbon
Qatar	the State of Qatar	(of) Qatar	Doha
Republic of Korea (the)	the Republic of Korea	of the Republic of Korea	Seoul
Republic of Moldova (the)	the Republic of Moldova	Moldovan	Chisinau
Romania	Romania	Romanian	Bucharest
Russian Federation (the)	the Russian Federation	of the Russian Federation, Russian	Moscow
Rwanda	the Rwandese Republic	Rwandan	Kigali
Saint Kitts and Nevis	Saint Kitts and Nevis	of Saint Kitts and Nevis	Basseterre
Saint Lucia	Saint Lucia	Saint Lucian	Castries
Saint Vincent and the Grenadines	Saint Vincent and the Grenadines	of Saint Vincent and the Grenadines	Kingstown
Samoa	the Independent State of Samoa	Samoan	Apia
San Marino	the Republic of San Marino	(of) San Marino	San Marino
Sao Tome and Principe	the Democratic Republic of Sao Tome and Principe	of Sao Tome and Principe	Sao Tome
Saudi Arabia	the Kingdom of Saudi Arabia	Saudi Arabian	Riyadh
Senegal	the Republic of Senegal	Senegalese	Dakar
Seychelles	the Republic of Seychelles	(of) Seychelles, a Seychellois	Victoria
Sierra Leone	the Republic of Sierra Leone	Sierra Leonean	Freetown
Singapore	the Republic of Singapore	Singaporean	
Slovakia	the Slovak Republic	Slovak	Bratislava
Slovenia	the Republic of Slovenia	Slovene	Ljubljana
Solomon Islands	Solomon Islands	(of) Solomon Islands	Honiara

Short name	Full name	Adjective/People	Capital city
Somalia	the Somali Democratic Republic	Somali	Mogadishu
South Africa	the Republic of South Africa	South African	Pretoria
Spain	the Kingdom of Spain	Spanish, a Spaniard	Madrid
Sri Lanka	the Democratic Socialist Republic of Sri Lanka	(of) Sri Lanka, Sri Lankan	Colombo
Sudan (the)	the Republic of the Sudan	Sudanese	Khartoum
Suriname	the Republic of Suriname	Surinamese	Paramaribo
Swaziland	the Kingdom of Swaziland	Swazi	Mbabane
Sweden	the Kingdom of Sweden	Swedish, a Swede	Stockholm
Switzerland	the Swiss Confederation	Swiss	Berne
Syrian Arab Republic (the)	the Syrian Arab Republic	Syrian	Damascus
<i>Tanzania: see United Republic of Tanzania (the)</i>			
Tajikistan	the Republic of Tajikistan	Tajik	Dushanbe
Thailand	the Kingdom of Thailand	Thai	Bangkok
The former Yugoslav Republic of Macedonia	The former Yugoslav Republic of Macedonia	of the former Yugoslav Republic of Macedonia	Skopje
Togo	the Togolese Republic	Togolese	Lomé
Tonga	the Kingdom of Tonga	Tongan	Nuku'alofa
Trinidad and Tobago	the Republic of Trinidad and Tobago	of Trinidad and Tobago	Port-of-Spain
Tunisia	the Republic of Tunisia	Tunisian	Tunis
Turkey	the Republic of Turkey	Turkish, a Turk	Ankara
Turkmenistan	Turkmenistan	Turkmen	Ashkhabat
Tuvalu	Tuvalu	Tuvaluan	Fongafale
Uganda	the Republic of Uganda	Ugandan	Kampala
Ukraine	Ukraine	Ukrainian	Kiev
United Arab Emirates (the)	the United Arab Emirates	of the United Arab Emirates	Abu Dhabi

Short name	Full name	Adjective/People	Capital city
United Kingdom of Great Britain and Northern Ireland (the) ¹	the United Kingdom of Great Britain and Northern Ireland	(of the) United Kingdom (of Great Britain and Northern Ireland), British	London
United Republic of Tanzania (the)	the United Republic of Tanzania	Tanzanian	Dodoma
United States of America (the) ²	the United States of America	(of the) United States (of America), American	Washington, DC
Uruguay	the Eastern Republic of Uruguay	Uruguayan	Montevideo
Uzbekistan	the Republic of Uzbekistan	Uzbek	Tashkent
Vanuatu	the Republic of Vanuatu	of Vanuatu	Vila
<i>Vatican City State: see Holy See</i>			
Venezuela	the Republic of Venezuela	Venezuelan	Caracas
Viet Nam	the Socialist Republic of Viet Nam	Vietnamese	Hanoi
Yemen	the Republic of Yemen	Yemeni, a Yemenite	Sana'a
Yugoslavia	Yugoslavia	Yugoslav	Belgrade
<i>Zaire: see Democratic Republic of the Congo (the)</i>			
Zambia	the Republic of Zambia	Zambian	Lusaka
Zimbabwe	the Republic of Zimbabwe	Zimbabwean	Harare

¹ The full name should be used on title pages, in signatures and in recording nominations, elections and votes. Once the full name has been mentioned or where space is limited (as in a table for instance), the following shorter form may be used: the United Kingdom.

² Once the full name has been used or where space is limited (as in a table for instance), the following shorter forms may be used: the United States, USA.

Annex 3

Place names

Alphabetical list by English/approved names

<i>English/approved name</i>	<i>Local/other name</i>	<i>English/approved name</i>	<i>Local/other name</i>
Abu Dhabi	Abu Zabi	Cairo	Al-Qahirah
Addis Ababa	Addis Abeba	Cambodia*	Kampuchea
Albania*	Shqipni, Shqipri, Shqipëri	Cameroon*	Cameroun
Alexandria	Al-Iskandariyah	Chad*	Tchad
Algeria*	Algérie	Cologne	Köln
Almaty	Alma-Ata	Conakry	Konakri
Antananarivo	Tananarivo	Copenhagen	København
Antwerp	Antwerpen, Anvers	Corsica	Corse
Athens	Athinai	Côte d'Ivoire*	Ivory Coast (the)
Azerbaijan*	Azerbaidzhan	Cracow	Krakow
		Cyprus*	Kıbrıs, Kypros
Baghdad	Bagdad	Damascus	Ash Sham, Damas, Dimash
Bahrain*	Bahrein	Dar es Salaam	Daressalam
Bangkok	Krung Thep	Dhaka	Dacca
Bangui	Bangi	Djibouti*	Jibuti
Basel	Bâle, Basle	Doha	Ad-Dawhah, Biha
Beijing	Peking	Estonia*	Eesti, Esthonia
Beirut	Bayrut, Beyrouth	Finland*	Suomi
Belarus*	Byelorussia, Belorussia	Florence	Firenze
Belgrade	Beograd	Gabon*	Gaboon, Gabun
Belize*	Belice	Gdansk	Danzig
Berne	Bern	Geneva	Genève, Genf
Bhutan*	Druk-yul	Genoa	Genova
Bosnia & Herzegovina*	Bosna & Hercegovina	Ghana*	Gana
Brazil*	Brasil	Ghent	Gand, Gent
Bruges	Brugge	Gothenburg	Göteborg
Brunswick	Braunschweig	Guangzhou	Canton
Brussels	Brussel, Bruxelles	Guinea*	Guinée(e)
Bucharest	Bucuresti		
Bulgaria*	Bulgariya		

* WHO Member States. For full, official names, see Annex 2.

<i>English/approved name</i>	<i>Local/other name</i>	<i>English/approved name</i>	<i>Local/other name</i>
Hague (The)	Den Haag, ’s Gravenhage	Lvov	Lwów
Haifa	Kaiffa, Khaifa	Lyon	Lyons
Haiti*	Haïti	Manama	Al-Manamah
Hanover	Hannover	Mantua	Mantova
Hanoi	Hanoi	Marrakesh	Marakesh, Marrakech
Havana	Habana (La)	Marseilles	Marseille
Helsinki	Helsingfors	Mauritania*	Mauritanie
Hungary*	Magyarország	Mauritius*	Ile Maurice
Iceland*	Island	Mecca	Makkah, Mekka
Iraq*	Irak	Milan	Milano
Ireland*	Eire	Mogadishu	Mogadiscio, Mogadisho
Jaffa	Yafa, Yafo	Morocco*	Al-Maghribiyah, Al-Mamlakah
Jakarta	Djakarta	Moscow	Moskva
Kathmandu	Katmandu, Khatmandu	Mozambique*	Moçambique
Kazakhstan*	Kazakstan	Munich	München
Kenya*	Kenia	Muscat	Maskat, Masqat
Khartoum	Khartum	Myanmar*	Burma
Kiev	Kiyev, Kiyiv	Naples	Napoli
Kuwait*	Koweit, Kuwait	N'Djamena	Fort-Lamy
Kyrgyzstan*	Kirghizia, Kirgizia	Netherlands (the)*	Holland
Latvia*	Latvija, Lettland	Norway*	Norge
Lebanon*	Liban	Nuremberg	Nürnberg
Leghorn	Livorno	Ostend	Oostende, Ostende
Libya*	Libia	Padua	Padova
Liège	Luik	Palau*	Belau, Belew
Lisbon	Lisboa	Phnom Penh	Pnompenh
Ljubljana	Lyublyana	Port Said	Bur Sa'id
Louvain	Leuven	Prague	Praha
Luanda	Loanda	Qatar*	Katar
Lucerne	Luzern	Reykjavik	Reikjavik
Luxembourg*	Luxemburg	Riyadh	Er Riad, Riad
Luxor	Al-Aqsar, Al-Qusur, Al-Uqsor		

* WHO Member States. For full, official names, see Annex 2.

<i>English/approved name</i>	<i>Local/other name</i>	<i>English/approved name</i>	<i>Local/other name</i>
Romania*	Roumania, Rumania	Thimphu	Tashi Chho Dzong, Thimbu
Rome	Roma	Tirana	Tiranë
Rwanda*	Ruanda	Tunisia*	Tunisie
Saint Kitts and Nevis*	Saint Christopher and Nevis	Turin	Torino
Salonica	Thessaloniki	Turkey*	Türkiye
Samarkand	Samarqand	Ulaanbaatar	Ulan Bator
Sana'a	Sanaa	Valletta	Valetta
Sao Tome*	Sao Thomé, Sao Tomé	Venice	Venezia
Saragossa	Zaragoza	Vienna	Wien
Sarajevo	Serajevo	Viet Nam*	Vietnam, Viet-Nam
Sardinia	Sardegna	Vilnius	Vilna, Vilno, Wilna, Wilno
Seoul	Kyongsong	Warsaw	Warszawa
Seville	Sevilla	Wroclaw	Breslau
Skopje	Skoplje	Yangon	Rangoon
Sofia	Sofiya, Sophia	Yaoundé	Yaunde
Sudan*	Soudan	Yerevan	Erevan, Erivan
Suriname*	Surinam	Yugoslavia*	Jugoslavia, Jugoslavija
Sweden*	Sverige	Zagreb	Zágráb
Szczecin	Stettin	Zurich	Zürich
Tajikistan*	Tadzhikistan		
Tallinn	Talin		
Tbilisi	Tiflis		
Teheran	Tehran		

* WHO Member States. For full, official names, see Annex 2.

Alphabetical list by local/other names

<i>Local/other name</i>	<i>English/approved name</i>	<i>Local/other name</i>	<i>English/approved name</i>
Abu Zabi	Abu Dhabi	Brussel, Bruxelles	Brussels
Ad-Dawhah	Doha	Bucuresti	Bucharest
Addis Abeba	Addis Ababa	Bulgariya	Bulgaria*
Al-Aqsur	Luxor	Burma	Myanmar*
Algérie	Algeria*	Bur Sa'id	Port Said
Al-Iskandariyah	Alexandria	Byelorussia	Belarus*
Al-Maghribiyah,	Morocco*		
Al-Mamlakah		Cameroun	Cameroon*
Al-Manamah	Manama	Canton	Guangzhou
Alma-Ata	Almaty	Corse	Corsica
Al-Qahirah	Cairo		
Al-Qusur,	Luxor	Dacca	Dhaka
Al-Uqsor		Damas	Damascus
Antwerpen,	Antwerp	Danzig	Gdansk
Anvers		Darressalam	Dar es Salaam
Ash Sham	Damascus	Den Haag	The Hague
Athinai	Athens	Dimash	Damascus
Azerbaidzhan	Azerbaijan*	Djakarta	Jakarta
		Druk-yul	Bhutan*
Bagdad	Baghdad		
Bahrein	Bahrain*	Eesti	Estonia*
Bâle	Basel	Eire	Ireland*
Bangi	Bangui	Erevan, Erivan	Yerevan
Basle	Basel	Er Riad	Riyadh
Bayrut	Beirut	Esthonia	Estonia*
Belau, Belew	Palau*		
Belice	Belize*	Firenze	Florence
Belorussia	Belarus*	Fort-Lamy	N'Djamena
Beograd	Belgrade		
Bern	Berne	Gaboon, Gabun	Gabon*
Beyrouth	Beirut	Gana	Ghana*
Biha	Doha	Gand	Ghent
Bosna &	Bosnia &	Genève, Genf	Geneva
Hercegovina	Herzegovina*	Genova	Genoa
Brasil	Brazil*	Gent	Ghent
Braunschweig	Brunswick	Göteborg	Gothenburg
Breslau	Wroclaw	's Gravenhage	The Hague
Brugge	Bruges	Guiné(e)	Guinea*

* WHO Member States. For full, official names, see Annex 2.

<i>Local/other name</i>	<i>English/approved name</i>	<i>Local/other name</i>	<i>English/approved name</i>
Habana (La)	Havana	Liban	Lebanon*
Haïti	Haiti*	Libia	Libya*
Hannover	Hanover	Lisboa	Lisbon
Hanoï	Hanoi	Livorno	Leghorn
Helsingfors	Helsinki	Loanda	Luanda
Holland	Netherlands (the)*	Luik	Liège
		Luxemburg	Luxembourg*
Ile Maurice	Mauritius*	Luzern	Lucerne
Irak	Iraq*	Lwów	Lvov
Island	Iceland*	Lyons	Lyon
Ivory Coast (the)	Côte d'Ivoire*	Lyublyana	Ljubljana
Jibuti	Djibouti*	Magyarország	Hungary*
Jugoslavia,	Yugoslavia*	Makkah	Mecca
Jugoslavija		Mantova	Mantua
		Marakesh,	Marrakesh
Kaiffa	Haifa	Marrakech	
Kampuchea	Cambodia*	Marseille	Marseilles
Katar	Qatar*	Maskat, Masqat	Muscat
Katmandu	Kathmandu	Mauritanie	Mauritania*
Kazakistan	Kazakhstan*	Mekka	Mecca
Kenia	Kenya*	Milano	Milan
Khaifa	Haifa	Moçambique	Mozambique*
Khartum	Khartoum	Mogadiscio,	Mogadishu
Khatmandu	Kathmandu	Mogadisho	
Kibris	Cyprus*	Moskva	Moscow
Kirghizia, Kirgizia	Kyrgyzstan*	München	Munich
Kiyev, Kiyiv	Kiev		
København	Copenhagen	Napoli	Naples
Köln	Cologne	Norge	Norway*
Konakri	Conakry	Nürnberg	Nuremberg
Koweit	Kuwait*		
Krakow	Cracow	Oostende,	Ostend
Krung Thep	Bangkok	Ostende	
Kuweit	Kuwait*		
Kyongsong	Seoul	Padova	Padua
Kypros	Cyprus*	Peking	Beijing
		Pnompenh	Phnom Penh
Latvija, Lettland	Latvia*	Praha	Prague
Leuven	Louvain		

* WHO Member States. For full, official names, see Annex 2.

<i>Local/other name</i>	<i>English/approved name</i>	<i>Local/other name</i>	<i>English/approved name</i>
Rangoon	Yangon	Tashi Chho	Thimphu
Reikjavik	Reykjavik	Dzong	
Riad	Riyadh	Tchad	Chad*
Roma	Rome	Tehran	Teheran
Roumania, Rumania	Romania*	Thessaloniki	Salonica
Ruanda	Rwanda*	Thimbu	Thimphu
		Tiflis	Tbilisi
		Tiranë	Tirana
Saint Christopher and Nevis	Saint Kitts and Nevis*	Torino	Turin
Samarqand	Samarkand	Tunisie	Tunisia*
Sanaa	Sana'a	Türkiye	Turkey*
Sao Thomé, Sao Tomé	Sao Tome*	Ulan Bator	Ulaanbaatar
Sardegna	Sardinia	Valetta	Valletta
Serajevo	Sarajevo	Venezia	Venice
Sevilla	Seville	Vietnam, Viet-Nam	Viet Nam*
Skoplje	Skopje	Vilna, Vilno	Vilnius
Sofiya, Sophia	Sofia		
Soudan	Sudan*	Warszawa	Warsaw
Shqipni, Shqipri, Shqipëri	Albania*	Wien	Vienna
Stettin	Szczecin	Wilna, Wilno	Vilnius
Suomi	Finland*	Yafa, Yafo	Jaffa
Surinam	Suriname*	Yaunde	Yaoundé
Sverige	Sweden*		
Tadzhikistan	Tajikistan*	Zágráb	Zagreb
Talin	Tallinn	Zaragoza	Saragossa
Tananarivo	Antananarivo	Zürich	Zurich

* WHO Member States. For full, official names, see Annex 2.

Annex 4

Terms used for manuscript mark-up and proof correction (English/French)

English	French	English	French
Typography			
typeface	caractère	in 2 columns	sur 2 colonnes
italic	italique	landscape layout	à l'italienne
roman	romain	not justified	en drapeau
bold	gras	full width	pleine justification
medium	mi-gras	on a fold-out page	en dépliant
light	léger	tint	trame
small capitals	petites capitales	screened	tramé
(large) capitals	capitales	Signs, symbols, etc.	
lower case	bas de casse (b.d.c.)	dash, em rule	tiret
diphthong	lettres accouplées	en rule [does not really exist in French]	1/2 moins
in smaller type	plus petit corps	hyphen	trait d'union
8pt, 10pt	c.8, c.10	minus sign	moins
Manuscript/proof mark-up		plus sign	signe plus
page make-up	mise en page	division sign	symbole de division
cover	couverture	multiplication sign	signe de multiplication
spine	dos	[note possible confusion with raised point]	
title page	page titre	raised point	point centré (signe de multiplication)
half-title page	page faux-titre	[medial multiplication point, water of crystallization]	
verso, left-hand page	verso	solidus	barre oblique, barre de fraction
recto, right-hand page	recto	(thin) rule	filet (maigre)
galley proof	épreuve en placard	subscript	inférieur
page proofs	épreuves en pages	superscript	supérieur
ozalid	ozalide	dotted line	ligne pointillée
blank page	page blanche	brace	accolade
title, heading	titre	parenthesis	parenthèse
running head	titre courant		
paragraph	alinéa		
footnote	note en bas de la page		

English	French	English	French
square bracket	crochet	add	ajouter
ellipsis [...]	points de suspension, ellipse	insert	insérer
thin/fine space	espace mi-fin	centre	centrer
prime sign	prime	transpose	transposer
bullet	gros point	align	aligner
asterisk	astérisque	take back [line, word, etc.]	passer à la ligne supérieure, à la ligne précédente
dagger	croix mortuaire	take over	à la ligne suivante
omission	bourdon	run on	faire suivre
Instructions		increase space	blanc à augmenter
indent	renforcer	reduce space	blanc à diminuer
to be typeset	composer	stet	bon
delete	supprimer	OK to print, OK for press	bon à tirer
close up	rapprocher		